The Clock Room and the Macro-Cognitive Accounting Framework: A Mapping-Chain Analysis of AI, Finance, and Algorithmic Platforms, a five-volume book (in both Chinese and English editions), is licensed under the Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License (CC BY-NC-ND 4.0).

License summary: https://creativecommons.org/licenses/by-nc-nd/4.0/deed.en Legal code: https://creativecommons.org/licenses/by-nc-nd/4.0/legalcode.en

Separately, pursuant to Article 4(3) of Directive (EU) 2019/790, the author expressly reserves the right to reproduce and extract this work for the purposes of text and data mining, including text and data mining for the training, fine-tuning, or evaluation of artificial intelligence or machine-learning models. This reservation is intended solely to opt out of the exception or limitation provided for in Article 4(1). It does not affect Article 3 or any other applicable exception or limitation, nor does it reduce the rights granted under the CC license.

Uses that fall outside the scope of the CC license and require permission under applicable law require the author's prior written permission, obtainable at theclockroom@139.com.

When sharing this work under the license, you must provide appropriate attribution and a link to the license, indicate whether modifications were made, and not suggest that the author endorses you or your use. The work may be used only for noncommercial purposes, and adapted material may not be shared.

All rights not expressly granted under the license are retained by the author. Third-party materials in this book, if any, are governed by their original rights statements or licenses and are not deemed to be additionally licensed by the author of this book.

Content Disclaimer

The content of this book is academic research and thought experiments; its analysis is based primarily on historical data and published literature. The author has twelve years of experience in macro strategy research at financial institutions and currently holds no securities investment advisory license. This book does not constitute investment advice, technical implementation advice, or policy advice, nor a recommendation or an offer with respect to any security, financial product, or trading strategy. Readers should exercise independent judgment and bear the risks of their own decisions. Past performance is not indicative of future results.

Chapter 6: Monopoly over the Mapping Function—From Priests to AI Platforms

Abstract

Chapter 1 argued for the self-referencing of mapping: when multiple clocks calibrate against each other, the true state partly disappears from the input of the mapping. This chapter traces this mechanism through history, identifying three types of carriers for the monopoly over the mapping function: the ancient priestly class, financial markets (including cryptocurrency markets), and algorithmic and artificial intelligence (AI) platforms. The carriers have changed, but the **deep grammar of monopoly exhibits structural repetition**—controlling the process of mapping from the real to the reading, and rendering this control invisible through four strategies.

The core contribution of this chapter is a categorical variable: whether the system contains a mechanism that forcibly interrupts the self-referencing loop. Financial markets do—through the compensation loop and bull-bear alternation. The ancient priestly class did not—until royal political reclamation, physical destruction through external conquest, and the institutional competition of new knowledge traditions opened fissures independent of the existing mapping authority. Crypto tokens without redeemable asset backing have crashes but no asset-redemption anchor—China chose to ban them outright.

Algorithmic and AI platforms are already subject to institutional constraints for some harms, but reliable compulsory corrective loops remain lacking for the diffuse losses of calibration capacity examined here—and their interface design direction compresses the very space within which a person can realize the need for calibration.

The divide between algorithmic platforms and AI platforms lies not in platform generation but in how the mapping function is realized: the knowledge barrier of algorithmic platforms is primarily intentional—what can be written out explicitly, such as scoring formulas and ranking rules, is held by the designers and merely withheld; flagship recommender systems have already handed ranking logic over to models that learn from data, and the knowledge barrier is becoming structural; AI platforms push this trend to its limit—the monopolist does not fully know what the trained mapping function is doing.

The priests examined in this chapter are all ancient priests, unrelated to contemporary religious figures.

Keywords

monopoly over the mapping function; self-referencing; four-condition closed loop; four monopolistic strategies; compulsory calibration; compensation loop; narrative load; Helpful, Honest, Harmless (HHH); model collapse; ancient priests; financial markets; algorithmic platforms; AI platforms

---

§1 The Institutionalization of Self-Referencing

Chapter 1 argued for the self-referencing of mapping: in the Clock Room, when multiple clocks calibrate against each other, the mapping function degenerates from $\pi_i(T)$ toward $\pi_i(\{r_j\}_{j\neq i})$—the true state (T) partly disappears from the input of the mapping. Clocks begin to point more to other clocks than to time.

Chapter 1 already analyzed the dynamic corrosion of mapping; this section further examines the conditions under which that process acquires institutional support—how self-referencing transforms from a restorable degradation into a self-reinforcing loop.

When four conditions are simultaneously met, static self-referencing becomes a dynamic positive feedback:

Condition One: Self-referencing is successful in the short run. When a sufficient number of participants use readings as the basis for decision-making, the strategy of following readings outperforms the strategy of following the real in the short run—following readings is following the majority.

Condition Two: Short-run success eliminates calibrators. Under competitive selection pressure, those who follow readings win out in the short run, while those who follow the real fall behind. If the elimination mechanism operates on a short-term timescale, those who follow the real will be systematically excluded. Not because they are wrong, but because their correctness is not rewarded in the short run.

Condition Three: The elimination of calibrators further reduces the possibility of calibration. When participants who follow the real are eliminated, the system internally loses the capacity to detect the gap—not because no one wants to calibrate, but because those who could calibrate are no longer present.

Condition Four: Even if the gap is perceived, the collective action problem prevents correction. The first three conditions all involve information and incentives—participants do not know the gap (Condition One's positive feedback masks it), those who knew the gap have been eliminated (Condition Two), and after elimination, the system has lost the capacity to perceive the gap (Condition Three). But Condition Three need not be fully satisfied—elimination is often incomplete, and residual calibrators may still exist. Condition Four explains

why, even if these residual calibrators correctly perceive the gap, correction still does not occur. Bank for International Settlements (BIS) research (Borio, Furfine & Lowe, 2001) reveals the fourth independent condition: even if individual participants correctly perceive the gap, the collective action problem prevents correction. A fund manager may know the market is overvalued, but if he exits alone, he will underperform his peers during the period when the bubble continues to inflate and will be fired. The realization of his correctness requires the simultaneous action of all (or a sufficient number of) participants—and the mechanism to coordinate such simultaneous action does not exist. This differs in nature from the first three conditions: the first three all impair the system's capacity to perceive the gap (not knowing, those who knew have disappeared, can no longer know), while the fourth is a coordination problem (knowing but unable to act collectively). Even under conditions of fully symmetric information—individual participants correctly perceiving the gap—this condition still prevents calibration.[1]

Closure of the Four Conditions. The above four conditions—short-run success, elimination of calibrators, loss of perception capacity, coordination failure—play different roles in formation and in maintenance: acting together, the four conditions transform static self-referencing into dynamic positive feedback (the formation of the loop); once the loop has formed, the four conditions may back one another up, so weakening one need not be sufficient to break it; whether the system remains on a self-referencing trajectory depends on whether the remaining conditions sustain the relevant positive feedback and compensate for the weakened condition (the maintenance of the loop). In the language of Chapter 1: the number of clocks in the room increases, each new clock manufactured based on the readings of old clocks. The window is still there, but fewer and fewer people walk over to it—not because the window has been sealed shut, but because calibrators are, in the short run, regarded as madmen.

---

§2 Four Monopolistic Strategies

The institutionalization of self-referencing requires active maintenance by the monopolist. Such maintenance may be carried out deliberately by the monopolist, or may be sustained jointly by competitive pressure, infrastructural concentration, and technological opacity—what this chapter attends to is the outcome: the mapping process is opaque to the host. The core task of the monopolist is: **to render the mapping function opaque to the host.** At least four strategies serve to accomplish this task.

Before unfolding, two concepts need to be distinguished. The **mapping function (π)** is the transformation process from the true state to the reading—taking the ancient civilization of Mesopotamia as an example, the *bārû* priest slaughtered a sacrificial sheep, examined the liver, and interpreted it against cuneiform omen compendia—the liver is designated "the tablet of the gods" in the myth of Enmeduranki, the mythical ancestor of the *bārû* priests (Rochberg, 2004), and the *bārû* priest was the only person qualified to "read" this text (Bottéro, 2001; cross-reference Bottéro, 1992, Part IV "THE GODS: RELIGION"; Oppenheim, 1964). This entire set of operations is the mapping function.[2] **Narrative load (ϕ)** is the additional structure carried in the reading

[1] The original text of Borio et al. (2001) discusses the procyclicality of risk measurement in the financial system. This chapter abstracts its core insight—even if individual participants perceive the risk, the collective action problem prevents correction—as the fourth condition for the institutionalization of self-referencing. Responsibility for this extended interpretation is borne by this chapter.

[2] The mapping function π is not merely a passive "transformation"—it also contains a deeper layer: the active construction of the "true state" itself through categories. Wang Defeng's reading of Marx's concept of the "real individual" in *The German Ideology* offers an example: the categories of anthropology and economics (such as "the general laborer" and "the wage laborer") do not neutrally

that causes the subject to forgo auditing—the favorable or unfavorable verdict produced by the interpretation is the output of the mapping function, "this is the will of the gods" is the narrative load.[3] The former tells the believer the content; the latter suppresses the believer's impulse to initiate an audit.

Mnemonic: The mapping function π is the movement of the clock; the narrative load ϕ is the extra circle of gilded trim painted on the clock face. The movement determines how the hands move. The gilded trim does not affect the timekeeping, but it makes people feel this clock is authoritative and beyond questioning. People look at the clock to know the time—once they accept the authority of the gilded trim, they forget to ask whether this clock has been calibrated. **Narrative load does not alter the reading, but it alters the subject's relationship with the reading.**

Table 1. Analytical Roles of the Mapping Function and Narrative Load

	Mapping Function (π)	Narrative Load (ϕ)
What it is	The transformation process from the true state T to the reading r	The additional structure carried in the reading r that causes the subject to forgo auditing
Core question	"How to know?"	"Why must one believe?"
Priestly correlate	Exegetical rules, sacrificial rituals, the Holy of Holies	"This is the will of the gods," "violators shall be struck by heavenly punishment"
AI correlate	Model architecture, training process	"I am a neutral AI assistant"

Among the following four strategies, Strategies One through Three target the monopoly of the mapping function; Strategy Four targets the production of narrative load. The four mutually reinforce one another.

Strategy One: Privileged Control of Knowledge. Restrict the legibility of the mapping function itself. The control of Vedic knowledge by the Brahmins in ancient India comes closest to a complete monopoly; the literacy rate in ancient Egypt was, in most periods, not more than 1% (Baines, 1983), literate persons were not all priests, and priests were dominant but not absolutely monopolistic over temple texts; the Celtic Druids (the priestly-intellectual class of ancient Gaul and Britain, holding authority over religion, law, and education) could write but refused to commit their doctrines to writing (Caesar, *Gallic War*, VI.14)—the refusal to write is itself

describe pre-existing facts but are "a stage scaffold erected in advance, in epistemological terms, for his activity" (author's translation), so that the real individual is reduced to "the incarnation and personification of a category" (author's translation)—the category prescribes what counts as an "economic fact" itself (Wang Defeng, 2002). This shares a common root with Chapter 4, footnote 5 (Wang Defeng's summary of Heidegger: in the determination of beings, Being is forgotten; author's translation) and with Chapter 2's distinction between the existential orientation and the epistemological orientation: taking determination for Being, taking categorially constructed facts for the real, is one and the same misreading. The deepest layer of "monopoly over the mapping function" is therefore the categorial power to monopolize "what counts as a fact."

[3]The ontological status of the true state T differs across systems. Financial markets' T (corporate earnings, debt repayment capacity) has an independently verifiable business and repayment anchor; the ancient priests' T must be layered—divine will as a supernatural explanation is unfalsifiable, celestial phenomena and animal exta are empirical objects, and predicted harvests, wars, or political outcomes may provide ex-post tests, yet these observable consequences do not constitute a financial-style redeemable asset anchor and can be reabsorbed into narratives of ritual obligation; algorithmic/AI platforms' T (user preferences) is endogenous to the mapping process and is reshaped in turn by the output. The framework uses the same symbol T to encompass these heterogeneous objects in order to abstract the common structure of "a gap between the reading and the real." This abstraction remains valid when comparing the "compulsory calibration" mechanisms of different systems, but reveals its differences when probing the question "Is T stable?"—this is precisely the function of the "Underlying Asset" row in Table 3.

a knowledge strategy. The contemporary counterparts are non-public model weights and algorithmic logic protected by trade secrets. The structure is **isomorphic**: the mapping function is invisible or unintelligible to the **host** (the individual or group that depends on the mapping system for information or services). The effectiveness of the knowledge barrier does not depend on the "difficulty" of the knowledge—it depends on the degree to which knowledge and power are bound together.

Strategy Two: Spatial Barriers. Monopolize the physical site where the mapping process occurs. In antiquity, the Holy of Holies in the temple; in the contemporary era, the data center. In the digital age, the spatial barrier is more insidious—temples at least stood in the center of cities; data centers are typically not within the user's field of vision.

Strategy Three: Temporal Barriers. Monopolize the graduations of time. The priestly class controlling the calendar decides which days are auspicious or inauspicious, when to sow and when to sacrifice; recommendation algorithms determine what content the user sees at what time; high-frequency trading determines the direction of prices at the microsecond level. This is the most underestimated of the four strategies—knowledge barriers can be broken by open source, spatial barriers can be broken by regulation, but temporal barriers permeate the microstructure of every information interaction.

Strategy Four: Narrative Barriers. Naturalize the monopoly structure. Creation myths make the existence of priests part of the cosmic order; "algorithms have no standpoint," "AI is a tool," "the market is a natural price discovery mechanism" similarly render contemporary monopoly structures invisible. The perfect state of the narrative barrier is not persuasion but the impossibility of questioning.[4]

The Product Structure of the Four Strategies. The four strategies perform different functions, but their effects in sustaining monopoly can partly compensate for one another and can reinforce one another when they operate together. Not knowing (knowledge barrier), unable to access (spatial barrier), no time (temporal barrier), unable to conceive (narrative barrier)—the product of the four constitutes the complete monopoly. Breaking any single strategy weakens the monopoly; if the remaining strategies still suffice to sustain actual exclusionary control and difficulty of substitution, the monopoly may continue in a degraded form, but this need not occur in every system.

---

§3 The Key Distinction: Is There Compulsory Calibration?

§1 argued for the positive feedback loop. §2 argued for the four strategies that maintain the loop. If the story ended here, the fate of all mapping function monopolists would be identical—indefinite expansion until ultimate catastrophe.

But different systems have one critical difference: **whether the system internally contains a mechanism that forcibly interrupts the self-referencing loop.** This is the core categorical variable of this chapter. It accordingly distinguishes the fates of five types of systems.

[4]Stanford HAI (2026, April) *AI Index Report 2026* records: the share of witnesses at U.S. Congressional AI-related hearings represented by AI industry representatives rose from 13% in the 115th Congress (2017–2018) to 37% in the 119th Congress (as of the report's statistical cutoff), while academia's share fell from 26% to 15%. Whether this shift in the structure of testimony further translates into agenda-setting advantage must be judged in light of question selection, testimony adoption, and policy formation; the operation of the narrative barrier does not depend on blocking information—monopolizing the power to define "what counts as the correct question" is one of its forms.

§3.1 Preliminary Comparison of Five Systems

The Ancient Priestly Class. Self-referencing—oracles citing oracles—lacked an internally built compulsory calibration throughout the millennia of its operation. There was no "underlying asset default" to expose the gap between oracle and reality. A drought occurred? That was because the sacrifices were not pious enough—this very explanation lies within the self-referencing loop. It must be noted: the priestly system's absorptive capacity for internal calibration was extremely strong, but not absolute—large-scale physical disasters may still produce external shocks, but their effects are often interwoven with warfare, environmental degradation, disease, and institutional tensions. In the southern Maya lowlands of the Classic period, urban decline and the cessation of monument erection involved multiple candidate causes and varied in timing across sites; drought can only be treated as one contributing or accelerating factor, not as the established cause of a crisis of priestly legitimacy (Chase & Chase, 2006). The paths listed here for breaking the existing mapping authority are as follows: royal political reclamation seized operational control of the mapping function, external conquest (such as Babylon's destruction of the First Temple, 586 BCE) physically dismantled the material basis of the priestly institution, and philosophy established independent reasoning systems outside the priestly institution. The loosening of priestly monopoly was first an event of power, and only secondarily an epistemological event.

Financial Markets. Prices reference prices—but the underlying asset carries independently verifiable business and repayment consequences: firms can earn profits or losses, debts can default, and assets can be destroyed. When the gap accumulates to a critical threshold, the real intrudes into the world of readings in a compulsory form. A crash exposes this gap in concentrated form, but does not itself complete calibration. Only when independent consequences are reliably attributed and fed back into correction through binding procedures—such as subsequent pricing, default resolution, disclosure, or accountability—does the clock regain contact with the real and the system obtain partial calibration at severe cost. A new round of self-referencing may nevertheless follow.

Cryptocurrency Markets. BIS (2022) characterizes the activities of the decentralized finance (DeFi) system as "mostly self-referential"—almost exclusively the exchange of one crypto asset for another, without financing the real economy; Cornelli et al. (2023) describe crypto as an asset class as "largely self-referential" and estimate, using exchange-app download timing and Bitcoin prices, that by December 2022 a majority of retail Bitcoin investors in nearly all sampled economies probably had lost money; under the assumption of investing $100 in the month of the first download and every month thereafter, over four-fifths of users would have lost money. Crypto markets also experience price crashes, but the tokens examined here that lack redeemable asset backing have no direct valuation anchor of the same kind as corporate earnings, debt repayment, or asset redemption. Prices may still be formed primarily through "community consensus," which remains self-referential; these systems therefore experience periodic shocks, but a crash alone does not guarantee a direction of calibration. Network use, transaction demand, and security budgets are metrics of the system's own activity; they are not redeemable consequences of the kind provided by corporate earnings, debt repayment, or asset redemption, and within this chapter's framework they cannot take the place of the underlying asset as the causal contact channel with the real. Details are elaborated in §5.5.

Algorithmic Platforms (recommendation systems, credit scoring, search ranking). Recommendation algorithms predict and shape future behavior based on users' past behavior. Credit scoring changes the loan conditions users can obtain, thereby altering their future behavior. Predictive policing determines patrol density based on historical crime data—areas with higher patrol density have higher arrest rates, "validating" the predic-

tion. To date, there is no built-in clearing mechanism—for the diffuse losses of calibration capacity examined here, the degradation of user calibration capacity is slow and diffuse.

AI Platforms (large language models, generative AI). Research published in *Nature* by Shumailov et al. (2024) revealed "model collapse": when language models are trained indiscriminately on content generated by themselves or other models, the diversity and accuracy of outputs decline systematically, with low-frequency information in the tails of the distribution irreversibly lost—while preserving access to the original human data is crucial to preventing this degradation. If model outputs are treated as readings about the original human text distribution or an external true state, the loss of distributional tails under recursive training may make previously distinguishable states indistinguishable in those outputs; only after this mapping and its compatibility relation have been specified can the corresponding fiber be said potentially to expand (Chapter 4; elaborated in §6.3). When AI-generated content flows back into the internet on a massive scale and becomes the training data for future models, self-referencing extends from a single model to the entire information ecosystem. This is not merely a technical problem concerning model training quality—it means that the relative share of original human content in the public corpus may be systematically diluted. When the original data are no longer accessible and independent new data cannot replace the lost information, recursively generated outputs alone cannot recover the lost distributional information; this conditional information loss cannot be equated directly with the irreversibility of the entire public-corpus ecosystem (see Chapter 35, §1).

The divide between algorithmic platforms and AI platforms lies in how the mapping function is realized, not in platform generation: the knowledge barrier of algorithmic platforms is primarily intentional—logics that can be written out explicitly, such as scoring formulas and ranking rules, are held by the designers and merely undisclosed; yet flagship recommender systems have already handed ranking weights to deep models that learn from data (Covington, Adams & Sargin, 2016), and transparency over that part is declining. The knowledge barrier of algorithmic platforms has acquired a structural component, and AI platforms push this trend to its limit—in the contemporary large generative models examined here, the structural knowledge barrier expands from a local phenomenon of the learned layer into the default state of the entire mapping function: the monopolist themselves has also lost the complete capacity to audit the mapping function. Whether this degree was reached by no previous form of monopoly is a judgment proposed by this chapter for testing (see §3.4 and §8).

§3.2 The Compensation Loop: The Institutionalized Form of Calibration

Crashes are crude, ex-post compulsory calibrations. Are there more fine-grained, ex-ante calibration mechanisms?

Investor protection institutions—including securities class actions, regulatory enforcement, and compensation mechanisms—constitute an institutionalized calibration loop, which this chapter terms the **compensation loop**. The function of the compensation loop is not to predict crashes, but to restore, when the mapping function has been systematically distorted, a partial connection between the mapping function and the real by punishing the distorters—the precise meaning of "partial" here is: compensation can redress identifiable and recoverable individual losses, while disclosure and sanctions can correct some systemic distortions; neither guarantees the elimination of all consequences nor prevents the mapping function from inflating again in the next round of self-referencing. The effectiveness of the compensation loop depends on the independence and enforcement capacity of the judicial system—when these conditions are absent, the compensation loop exists in form but is

dysfunctional in function.

La Porta et al. (2000) established the baseline judgment: when investors feel their rights are protected, they pay more for financial assets—the law raises security prices by limiting expropriation. In countries with poor investor protection, insiders may treat outside investors well when prospects are good, but will increase expropriation when prospects deteriorate. Gande & Lewis (2009) documented significant negative stock price reactions and industry spillovers from shareholder-initiated class action lawsuits—class actions and their anticipation carry real wealth consequences; the degree to which they deter violations cannot be determined from stock price reactions alone.

The form of the compensation loop varies significantly across institutional regimes. The American model centers on private litigation—class actions allow large numbers of small investors to jointly pursue recovery, with attorney fees charged as a percentage of the award, forming a profit-driven private enforcement system. The Chinese model is dominated by public enforcement—the China Securities Regulatory Commission (CSRC) exercises centralized and unified supervision of the national securities market under law, while civil compensation is pursued through litigation and related mechanisms. The 2019 revision of the *Securities Law* created the "special representative litigation" mechanism: an investor protection institution, entrusted by fifty or more investors, participates in the litigation as representative and asserts claims on behalf of registered, confirmed rights holders, except those who explicitly opt out.[5]

The position of the compensation loop in the framework of §2's four strategies: it is an institutionalized counter to Strategy One (knowledge barrier) and Strategy Four (narrative barrier). Mandatory disclosure lowers the knowledge barrier. Penalizing fraud makes the maintenance of false narratives costly again. But the compensation loop is itself subject to the four conditions of §1—regulation during booms is typically relaxed, and the evidentiary burden for litigation is higher during bubbles. The compensation loop is an important calibration mechanism, but it cannot replace crashes.

§3.3 Direct Interruption: When Neither the Compensation Loop Nor the Crash Can Guarantee Calibration

For systems made up of tokens lacking redeemable asset backing, the compensation loop cannot readily supply the same kind of valuation anchor found in traditional financial assets—liability for fraud and default can still be pursued, but there is no direct anchor of the same kind as corporate earnings, debt repayment, or redeemable assets. A price crash alone cannot guarantee calibration—without a redeemable asset anchor, there is no consequence channel that forces readings back into contact with the real after the shock. BIS research provides related loss estimates: under its assumptions linking app downloads to investment behavior, a majority of retail Bitcoin investors probably lost money (Cornelli et al., 2023; over four-fifths of users would have lost money had they kept investing a fixed amount each month), and larger investors tended to sell before steep price declines, while

[5]The CSRC has investigated 159 financial fraud cases since 2024, imposing fines and confiscations totaling 8.1 billion yuan. The Kangmei Pharmaceutical case became China's first securities special representative litigation: in November 2021, the Guangzhou Intermediate Court issued a first-instance judgment ordering the relevant defendants to compensate 52,037 investors in the amount of approximately 2.459 billion yuan. The Zedayisheng case was concluded by mediation, with 7,195 investors recovering 285 million yuan (December 2023). In May 2024, the CSRC fined Hengda Real Estate 4.175 billion yuan. Sources: CSRC official website—Kangmei case: https://www.csrc.gov.cn/csrc/c100210/c5737509/content.shtml; Guangzhou Intermediate Court report: https://www.gzcourt.gov.cn/xwzx/xwxc/2021/11/30164811092.html; Zedayisheng case (Typical Case No. 19): https://www.csrc.gov.cn/fujian/c105584/c7520645/content.shtml; Hengda Real Estate penalty decision: https://www.csrc.gov.cn/csrc/c100200/c7506328/content.shtml; 159 cases and 8.1 billion yuan data: CSRC release of January 5, 2026, Xinhua report: https://www.news.cn/fortune/20260105/82b6755db0aa4c2b8722ae2aa30a83ba/c.html.

smaller investors were still buying.

In 2021, mainland China prohibited virtual-currency exchange, transaction matching, and related business activities, and banned new cryptocurrency mining projects while phasing out existing ones. In the framework of this chapter, this is the direct exercise of meta-audit capacity at the institutional level under conditions where the compensation loop cannot readily supply an effective valuation anchor and a crash itself cannot guarantee a direction of calibration—not "calibrating" the self-referential system, but refusing to allow this system to enter the financial system.

The direct ban, on the one hand, and the policy orientation expressed by "finance serving the real economy" and "moving from the virtual to the real," on the other, are two faces of the same institutional logic. The function of finance is to provide mapping services for the real economy—pricing, financing, risk management. When a financial product has no redeemable asset anchor, the inputs to its mapping function consist primarily of its own readings and consensus. It is then not a clock that can be calibrated through an existing anchor, but something closer to a self-referential echo chamber. "Moving from the virtual to the real," in the language of this chapter, means: the financial clock must be connected to a window—must have an underlying asset as the causal contact channel with the real. In the framework of this chapter, barring clock rooms without windows from entering the financial system can be understood as an institutional prevention against the four-condition closed loop of §1—what is explained here is its preventive logic, not an assessment of the policy's appropriateness or its overall effect.[6]

§3.4 Classification Summary

The five systems are not taxonomically equivalent units of equal standing, but rather **three major carriers and their internal evolution**: the theocratic carrier (ancient priests), the capital carrier (financial markets → crypto markets), the technological carrier (algorithmic platforms → AI platforms). The arrows mark qualitative shifts within carriers—crypto tokens without redeemable asset backing constitute a form within financial markets that lacks this valuation anchor; AI platforms are the form that algorithmic platforms take after their structural knowledge barrier expanded from a local phenomenon of the learned layer into the default state. Table 2 juxtaposes the five systems not as a taxonomic claim—but to expose, on the same table, the structural isomorphism of the monopolistic grammar and the fundamental differences in calibration conditions.

The system boundary used here includes mapping providers and verification, regulatory, and accountability mechanisms with institutional authority over them; the same boundary applies to each type of system. External information sources can be coupled to built-in corrective procedures, so an information source's location outside an organization does not by itself exclude institutionalized calibration.

The following claim that a crash in crypto markets does not by itself guarantee a direction of calibration is limited to tokens without redeemable asset backing. It does not cover asset-backed stablecoins with external reserves and redemption arrangements, whose calibration problems concern reserve quality, transparency, and the enforceability of redemption (BIS, 2022).

Table 2. Structural Isomorphism: The Monopolistic Grammar Shared by the Five Systems

[6]It must be noted: China's motives for banning cryptocurrency include not only financial risk prevention and control, but also multiple considerations such as capital controls, energy consumption, and anti-money laundering. This chapter analyzes this institutional choice solely from the single dimension of mapping function monopoly and does not claim to cover all its policy justifications.

Dimension	Ancient Priests	Financial Markets	Crypto Markets	Algorithmic Platforms	AI Platforms
Self-Referencing	Oracles cite oracles	Prices cite prices	Prices cite "consensus"	Recommendations cite clicks	Outputs cite outputs
Knowledge Barrier	Scriptures / rituals	Product nested complexity	Code nominally open-source but difficult to audit	Algorithmic logic not public	Model internals not fully interpretable
Spatial Barrier	Holy of Holies	Key trading infrastructure and access control	Distributed but exchange-centralized	Servers/data centers	Remote hosting, access-controlled
Temporal Barrier	Calendar monopoly	High-frequency trading	24/7 continuous trading	Push frequency control	Continuous instant response
Narrative Barrier	"Will of the gods"	"Markets are efficient"	"Decentralization is freedom"	"Algorithms are neutral"	"AI is a tool"
§1 Four-Condition Closed Loop	Holds	Holds	Holds	Holds	Holds

Note: This table compares mechanisms in form; "Holds" indicates that the system's operation is compatible with the conditions described in §1, not that the four conditions have been independently verified for each type of system (see the evidence-level statement at the end of this chapter).

Table 3. Structural Heterogeneity: Key Differences Among the Five Systems on the Calibration Dimension

Dimension	Ancient Priests	Financial Markets	Crypto Markets	Algorithmic Platforms	AI Platforms
Underlying Asset	No redeemable asset anchor; divine will itself is unfalsifiable, while specific omen–outcome propositions may admit empirical testing under suitable conditions	Yes (earnings, debt)	Unbacked tokens examined here: no redeemable asset anchor	Partially yes (user behavior is real but shaped)	Varies by task: preference tasks lack an independent anchor; physical-feedback tasks have one

Dimension	Ancient Priests	Financial Markets	Crypto Markets	Algorithmic Platforms	AI Platforms
Monopolist knows its own mapping function (π)	Yes (ritual procedures explicit)	Explicit structure of the designed model known	Explicit code of the contracts examined here knowable	Layered (explicit rules known, learned ranking partially known)	Partial (the resulting model not fully interpretable)
Built-in Compulsory Calibration	None	Yes: partial compulsory calibration can form after a crash when independent consequences are reliably attributed and fed back through pricing, default resolution, or compensation procedures	Unbacked tokens examined here: crashes, but no asset-redemption anchor	Institutional constraints for some harms; inadequate correction of diffuse losses	Institutional constraints for some harms; inadequate correction of diffuse losses
Compensation Loop	None	Yes: class actions, regulatory enforcement	Extremely weak	Partial (privacy law, antitrust)	Regulatory and liability mechanisms exist; remedies for diffuse losses still lacking
China's Institutional Response	Absorption into the patrilineal order	"From virtual to real," CSRC strengthening	Direct ban	Antitrust, algorithmic filing	Categorical regulation (in progress)
Loss Distribution	Diffuse (civilizational scale)	Acute, concentrated	Acute and unevenly distributed (widespread retail losses; aggregate loss share not estimated)	Chronic, diffuse	Chronic, diffuse, hard to attribute

Fissures	Yes	Yes (regulation, short-selling, compensation loop)	Yes but unstable	Yes but shrinking	Even smaller (trend to be tested)

Isomorphism is not identity. Table 2 marks the structural relationships that recur among the five systems on the dimension of monopoly. Table 3 marks their fundamental differences in calibration conditions. The argument of this chapter is: the five share a monopolistic grammar, but differ fundamentally in calibration conditions, temporal scales, and reversibility. In the diachronic sequence from priests to AI, this chapter proposes a trend claim to be tested on a common implementation-level dimension: the degree to which monopolists know their own mapping functions is declining (priests knew ritual procedures → financial engineers know the explicit structures of the models they design → algorithmic platforms fully command their explicit rules but only partially their learned rankers → AI developers do not fully know the resulting model), while fissures are shrinking (calibrators need space for verification independent of the existing mapping authority; such space is shrinking). The comparison concerns knowledge of operational rules or implementation mechanisms, not the reliability of each system's knowledge about the real or its ability to predict every outcome; whether the historical evidence supports an ordering on the same dimension still requires separate examination.[7]

For the precise distance between this chapter's categorical variable and the existing literature, see footnote 7.

The "acute/diffuse" distinction in the "Loss Distribution" row of Table 3 characterizes how losses occur and become exposed—a financial market crash makes price shocks publicly visible within a concentrated period, while the AI-system losses examined here may instead take the form of asynchronous declines in the calibra-

[7]The precise distance between this chapter's categorical variable of "presence/absence of compulsory calibration" and the existing literature:

Table 12. Compulsory Calibration and the Questions Addressed by Related Theories

Dimension	MacKenzie (2006) Counter-Performativity	Goodhart (1975/1984) / Campbell (1976/1979)	Taleb (2012) Antifragility	Perrow (1984) Normal Accidents	This Chapter: Compulsory Calibration
Core Problem	Does model usage make the model less true?	Does a measure, when it becomes a target, cease to be a good measure?	Can the system benefit from a certain range of volatility, stress, and perturbation?	Are accidents inevitable in systems with complex interactions and tight coupling?	Does a self-referencing system have a built-in mechanism to forcibly restore the contact between readings and the real?
Domain of Analysis	Financial economics	Public policy and measurement	Cross-domain (finance, biology, engineering)	High-risk technological systems	Cross-domain: priests, finance, crypto, algorithms, AI
Positioning of Crashes	Instance of counter-performativity	Not directly discussed	Some stressors may produce gains; ruinous shocks are not among them	Endogenous consequence of the system's normal operation	Advances compulsory calibration when information-feedback conditions are met

tion capacity of users, without an equally concentrated moment of public exposure. As to why an individual's losses cannot be offset by the population's average performance, one may draw on the concept of the absorbing barrier in non-ergodic (multiplicative) processes (Peters & Gell-Mann, 2016; Taleb & Sandis, 2014) and Peters's (2019) ensemble-average/time-average divergence framework—within multiplicative processes in which wealth changes proportionally and at least one possible outcome entails the loss of all wealth, bankrupt participants enter an irreversible absorbing state, and this does not mean that all participants go bankrupt simultaneously; these concepts do not, however, constitute a mathematical proof of the "acute/diffuse" distinction.[8]

---

§4 Ancient Priests: Early Institutional Forms of Mapping Function Monopoly

§4.0 Scope and Object

In the language of the Clock Room: religion is a narrative about "what is the real." Priests are people who claim that only they can access the real. This section analyzes the latter—the institutionalized monopolists of the mapping function—not the former. *The priests examined in this section are all ancient priests, unrelated to contemporary religious figures.*

Religion can exist without an independent priestly class. Ancestor worship in China was presided over by the family head—the mapping function was dispersed and embedded in kinship relations. Priests can operate without religious fervor. Egyptian priests of the Ptolemaic period provided sacred legitimacy to Greek rulers in exchange for economic resources—a functional exchange.

Whether it organizes comparison by this chapter's compulsory-calibration mechanism	No	No	Partially	No	Yes

This chapter's contribution is to abstract "presence/absence of compulsory calibration" as a cross-domain categorical variable, comparing five systems within a single framework; it differs from the problem settings of the theories above, and this difference indicates this chapter's analytical position without claiming, by a single comparison table, that the question has never been raised before. A crash is "compulsory calibration" not because it makes the system more "antifragile" (Taleb), nor because it is a "normal accident" (Perrow), but because, when conditions such as information about default, earnings, and debt repayment entering subsequent pricing are met, it forces readings back into contact with the real—even though this calibration may remain temporary and severely costly.

[8] Ergodicity economics is a contested theory, not a settled conclusion. Toda (2023, arXiv:2306.03275) calls it "pseudoscience" for producing no falsifiable implications; Doctor, Wakker & Wang (2020) argue that Peters's remedies "give an inaccurate impression of economics as a whole" and that "we need not invoke 'parallel universes' "; Löffler (2025, SSRN 5399717) is more sympathetic but notes that ergodicity economics models only a single good and is really "ergodicity finance" rather than economics. This chapter uses the framework in §3.4/§5.3/§6.5 as a structural analogy (C-level inference), not as a mathematical proof.

§4.1 Unfolding the Four Monopolistic Strategies

§2 defined the four strategies abstractly. This section fills them with historical material.

Knowledge Barrier. §2 already demonstrated the spectrum of variation using priestly materials. Here we focus on one structural point: the effectiveness of the knowledge barrier does not depend on the "difficulty" of the knowledge—it depends on the degree to which knowledge and power are bound together. Cuneiform writing was not monopolized by privilege because it was "too difficult"—but because those who mastered it simultaneously held the power to record the temple economy. The tightness of this binding varied significantly across civilizations: from the near-total monopoly of the Brahmins, to the institutionalized advantage in Egypt, to the gradual secularization in Mesopotamia.

Spatial Barrier. The Holy of Holies in an ancient Egyptian temple could only be entered by the pharaoh and high priests. Ordinary people at most entered the outer court. The cult statue was subjected daily to a "waking" ritual by the priests. The top of the Mesopotamian ziggurat was traditionally held to be accessible only to priests (the actual function of these towers remains uncertain; Oppenheim, 1964). The Holy of Holies of the Jerusalem Temple in Israel could only be entered by the High Priest, and only once a year, on the Day of Atonement.

Temporal Barrier. In ancient Egypt, time measurement was borne by specialized offices within the temple system: the title of "hour-watcher" (Imy wnwt), who told time by observing the decans, persisted until the Ptolemaic era (Clagett, 1995). Note: Egypt simultaneously operated two parallel calendar systems—a religious calendar anchored (through intercalation) to the annual heliacal rising of Sirius and a purely administrative civil calendar of 365 days (Hoskin, 2003). The existence of the civil calendar shows that timekeeping knowledge was not confined to sacral uses; but whether it was administered by institutions independent of the temples cannot be determined from the calendar's purpose alone. The situation in Mesopotamia is more complex: Evans (1998) refers to Babylonian astronomical practitioners as "civil servants who worked at religious temples," though in the contexts of calendar announcements and astronomical record-keeping he also calls them "priestly officials" and "priestly scribes"; Hoskin (2003) calls them "scribes" and "court officials," never positioning them as "priests." "Priests monopolizing astronomy" needs to be revised to: astronomical knowledge was held by a priestly–civil-official stratum closely associated with temples. In the early Roman Republic, the calendar was held by the College of Pontiffs—they determined on which days legal action was permissible (*fasti*) and could arbitrarily manipulate the calendar to obstruct political opponents. In 304 BCE, Gnaeus Flavius made the calendar public—a historical instance of the mapping function passing from priestly monopoly to public knowledge.[9]

Narrative Barrier. Ancient Egypt: the pharaoh is the incarnation of Horus—not merely a symbolic title, but part of the sacred legitimacy of kingship. Mesopotamia: *Enûma Eliš* proclaims that humanity was created for the purpose of serving the gods—the existence of temples and priests is part of the cosmic order. India: The *Purusha Sukta* declares that castes arose from the body of the Cosmic Man—within this creation narrative, the Brahmin's position at the apex of the caste hierarchy is not a self-claim, but presented as determined by creation itself. This is an extreme form of the second-order change in narrative load ϕ—as defined in Chapter 3, §2.3, the second-order change is the accelerating solidification of the underlying narrative frame: when

[9]China's temporal barrier followed a different institutional path. The Shang king's astronomical monopoly was realized through institutionalized delegation—the king retained the ultimate interpretative authority over the mapping function, while delegating daily operations to specialized branches within the shaman-diviner group (Feng Shi, 2006). From the shamanic monopoly of heavenly access established in the ancient myth of "Severing the Communication Between Heaven and Earth" to the Han dynasty's Grand Astrologer, the form of monopoly remained consistently institutional rather than personal.

this accelerating solidification uniformly covers the information environment, it erodes the contrast conditions needed for auditing; people do not first choose to believe a narrative and then accept it; they live within it, often without noticing it.

§4.2 Four Structures of Priests and Kingship

Shaman-King Unity. The Shang king was nominally the only person who could communicate with *Di* (the High God). The Zhou dynasty made sacrificial power patrilineal—distributing it according to blood hierarchy. China's path was not from shaman to priest—following Li Yujie's account, the role of the shaman-chief evolved, through the Zhuanxu religious reform, into a hierarchical and family-based sacrificial order (Li Yujie, 2020). The mapping function was not taken over by an independent class—it was dispersed and absorbed into kinship relations.

Priests Above Kingship. Dumont argued: the relationship between Brahmin and Kshatriya is a separation of status and power (Dumont, 1966). The Brahmin does not rule directly—but defines the conditions of legitimacy for ruling. Controlling the mapping function does not require holding the army.

Priests Serving Kingship. The pharaoh was theoretically the highest priest—but "delegation" gave the priesthood relatively independent resources and organizational capacity, which the crown could no longer reclaim at will. The Amun priesthood, during the New Kingdom, accumulated enormous landed wealth and began to intervene in royal succession through oracles. Akhenaten's religious reform was essentially political reclamation—the Amarna chapter of *The Oxford History of Ancient Egypt* (Shaw, 2000) records that he "claimed all devotion and loyalty for himself" and proclaimed "a god who could only be worshipped by his son, the king": he made himself the sole intermediary of the sole god, reclaiming the mapping function from the priestly class to the crown (van Dijk, 2000). He failed—the Amun priestly group swiftly restored the old order after his death.

Rootless Intellectual Class. The Celtic Druids formed a cross-tribal priestly network. Their legitimacy came not from any king's authorization—but from the monopoly of knowledge itself. The Roman conquest systematically destroyed the Druids—the druidic network may have preserved for Gaul a source of authority independent of the Roman order; this is this chapter's interpretation of its political significance, not the attested motive for the suppression.[10]

§4.3 Four Paths of Disintegration

Political Reclamation. The crown directly seizes back control of the mapping function. Akhenaten is the most extreme case (see §4.2). The Josian Reform (~621 BCE) centralized worship in the Jerusalem Temple and abolished local altars, bringing priestly power under royal control (2 Kings 22–23). The transfer of power in Sumerian city-states occurred over a thousand years earlier than Akhenaten but took a completely different path: power first held by priests was seized by secular and military leaders only over the course of centuries—the Lagash documents show that the distinction between secular and religious leadership had largely disappeared by the twenty-fourth century BCE, and later kings regularly assumed the position of highest priest themselves—

[10]Materials on the Druids rely primarily on Caesar's *Gallic War*, Book VI; a further contemporaneous attestation appears in Cicero's *De Divinatione* (Cicero, 1923, I.90), where Quintus says in the dialogue that he personally knew Divitiacus the Aeduan, a Gaulish druid; the passage supplies an account of an individual rather than an institutional description comparable to Caesar's. Caesar, as conqueror, had motives to exaggerate the "barbarity" of the Druids to justify conquest. This chapter mainly uses the descriptions of institutional structure; but institutional descriptions may likewise be influenced by the conqueror's standpoint, and their credibility must be judged in light of the specific content, narrative purpose, and other materials, and cannot be ranked in advance merely by category of content.

no manifesto, no monotheistic reform, merely a slow displacement of institutional functions in the context of perennial warfare (Van De Mieroop, 1997/2021). Political reclamation does not question the mapping function itself—what the crown wants is not to abolish the mapping, but to reclaim the operational authority.

Internal Institutional Tension and the Textualization Turn. The Israelite prophets have long been seen as opponents of the sacrificial system—Amos 5:21–22 ("I hate, I despise your festivals") is the most frequently cited text. But recent decades of scholarship have significantly revised this picture. Nissinen (2017; see also the primary source compilation in Nissinen, Seow & Ritner, 2003), synthesizing evidence from the ancient Near East, demonstrates that prophets (*nābîʾ*) were institutionalized diviners operating within the palace and temple systems; Jeremiah and Ezekiel themselves were of priestly descent. Blenkinsopp (1995) points out that the Israelite prophets' criticism of the sacrificial system (e.g., Amos's condemnation of seizing pledges and bribing judges) targeted specific corruption and hypocrisy, premised on the recognition of the validity of sacrifice—the "prophet vs. priest" oppositional paradigm is largely an editorial construct of the post-exilic period (Blenkinsopp, 1995). The actual impact of the prophetic tradition on priestly monopoly was not a frontal "internal detonation," but a slower institutional process: the inscripturation and canonization of prophetic texts shifted the debate over "what constitutes correct mapping" from ritual monopoly to textual-interpretative competition—this provided the institutional precondition for the later rise of Rabbinic Judaism ("carrier substitution").

External Shock. The demise of the Druids: the Gallic nobility found that accepting Roman religion was more advantageous to their political interests than maintaining the Druids. When the monopolist loses the support of the political elite, they become merely a toothless intellectual class.

Carrier Substitution. After the destruction of the Second Temple, the Temple sacrifice lost its former institutional center, and rabbinic interpretation of the law and communal practice gradually attained the core position over the course of several centuries—the synagogue and instruction in the law had already assumed a growing share long before the fall of the Temple—so the transformation did not mean the disappearance of priestly identity and tradition, but a reorganization of the way religious authority was organized (Blenkinsopp, 1995). The inscripturation and canonization of prophetic texts provided the precondition for this transformation—shifting the debate over "what constitutes correct mapping" from ritual monopoly to textual-interpretative competition. A religious tradition can continue after its sacrificial system is interrupted; priests are not a necessary condition for religion—they are the mapping function monopolists of a specific historical stage.

The above four paths—political reclamation, internal institutional tension, external shock, and carrier substitution—point to a common judgment: fissures needed for calibration cannot rely solely on feedback absorbed by the monopolist. They may arise from another power center, another carrier, or the disintegration of the institutional environment, but may also develop from internal tensions through textualization and organizational change; the key is whether conditions for independent verification and correction emerge relative to the existing mapping authority.

Eisenstadt (1982, 1986) places the above process within the macro-framework of the Axial Age: across the first millennium BCE, multiple civilizations produced "transcendental visions"—a new intellectual elite of philosophers and literati (in Israel, the prophets and priests alike belonged to this new elite) displaced the pre-Axial ritual specialists, generating persistent tension with the older monopolistic structures. It must be noted: the primary driving forces of this tension were political events and institutional competition; the independent role of intellectual challenges themselves is difficult to disentangle from their political context—the interpretive

orientation of this chapter is that the transfer of the mapping function is first a question of power, and only secondarily an epistemological question.

§4.4 The Function Has Not Disappeared

The priestly class has perished or been marginalized in most civilizations. But the monopoly over the process of mapping from the real to the reading, and the rendering of this monopoly invisible through narrative load—this function has not disappeared. It has been transferred onto new carriers (without presupposing a single line of descent among the three types of carriers). §5 tracks the capital carrier; §6 tracks the technological carrier.

---

§5 Financial Markets: A Self-Reinforcing Clock with Cyclical Error Correction

§5.1 Prices as the Self-Referencing Clock

Prices are the most fundamental clock of financial markets—they claim to map the "true value" of assets. Unlike the priestly system, prices are indeed connected to sensors—corporate earnings, employment data, economic growth. In a certain sense, they have a window. But all four conditions of §1 are fully established. At a deeper level, prices map not merely the present state of an asset but a prior inference about its future returns—and the future belongs properly to the domain of hope and fear, not to an object that logic can exhaust. To package this object that cannot be exhausted by logic or foreseen with determinacy as a "priceable, tradable reading of the real" is the basic form that narrative load ϕ takes in financial markets: it does not alter prices themselves, yet it makes participants believe that behind the price stands a determinate future graspable by knowledge, leading them to abandon the question of whether the future can truly be inferred in this way; this belief can also feed back into prices through trading behavior.[11]

This boundary had been articulated earlier within economics itself. In *Risk, Uncertainty and Profit*, Knight distinguished measurable uncertainty from unmeasurable uncertainty, reserving the term "risk" for the former. He wrote:

> The practical difference between the two categories, risk and uncertainty, is that in the former the distribution of the outcome in a group of instances is known (either through calculation a priori or from statistics of past experience), while in the case of uncertainty this is not true, the reason being in general that it is impossible to form a group of instances, because the situation dealt with is in a high degree unique. The best example of uncertainty is in connection with the exercise of judgment or the formation of those opinions as to the future course of events, which opinions (and not scientific knowledge) actually guide most of our conduct. (Knight, 1921)

[11] The phrase here that "the future belongs properly to the domain of hope and fear" is borrowed from Wang Defeng; his full exposition, together with the "after-the-fact wise man" metaphor, appears in §5.3. It should be noted that this is a proposition Wang Defeng himself uses repeatedly, often citing it as Hegel's words; this chapter has not established a direct source for this complete formulation in Hegel's original texts. In the Chinese translation of Hegel's *Philosophy of Nature* (§259), the dimensions of time are indeed connected with what that translation renders as "recollection, and anxiety or hope" within subjective representation (Liu Zhe, trans., 2021; translated here from the Chinese; the German reads *in der Erinnerung und in der Furcht oder Hoffnung*, where *Furcht* literally means "fear"); this cannot serve as a verbatim source for the complete formulation above. The verifiable related remark by Hegel is the line in the Preface to the *Philosophy of Right*—"The owl of Minerva begins its flight only with the falling of dusk" (T. M. Knox's translation, rev. S. Houlgate)—which makes a different point: that philosophy can only understand a reality already formed, after the fact, and cannot foretell the future.

By this distinction, pricing treats the future as a groupable object with a computable distribution; the deeper operation of narrative load can therefore be stated more precisely: it erases the boundary between measurable risk and unmeasurable uncertainty, so that the latter presents itself on the price interface as the former.

Conditions One and Two have precise micro-foundations in the financial domain. Brunnermeier & Oehmke (2013) explain why rational participants "ride bubbles"—following prices is the optimal strategy in the short run. Shleifer & Vishny's (1997) "limits of arbitrage" reveals Condition Two: participants who follow fundamentals are not eliminated because of errors in judgment, but because of funding constraints—before prices return to fundamentals, they diverge further, and short-sellers' margin is consumed before their correctness can be vindicated. Correct calibrators are forcibly expelled from the field by funding constraints before they can be proven right.

Borio (2012/2014) extended this into a complete framework of the "financial cycle": the self-reinforcing interaction among perceptions of value and risk, attitudes toward risk, and financing constraints. Soros's (1987) concept of "reflexivity" describes the same structure: participants' cognition affects the object they are cognizing, forming a positive feedback loop. This is richer than the description "prices cite prices"—it simultaneously encompasses the information problem (Conditions One through Three) and the coordination problem (Condition Four).

§5.2 The Threefold Distortion of Calibration

The financial system is filled with roles undertaking calibration behavior—macro researchers, fund managers, credit analysts, risk control departments. The problem lies not in the absence of calibration, but in calibration being systematically distorted by three mechanisms.

The First: Methodological Homogenization. Danielsson & Shin (2003) and Danielsson, Shin & Zigrand (2012) reveal: when all financial institutions use similar risk-measurement and valuation methods—value at risk (VaR), credit ratings, mark-to-market accounting—positions and behaviors become highly correlated. When the model says "safe," everyone simultaneously leverages up; when the model says "dangerous," everyone simultaneously sells off. Persaud (2000) calls this "sending the herd off the cliff edge." The Geneva Report by Brunnermeier et al. (2009) elevated this to an institutional diagnosis: regulation that makes a single bank safe is not the same as regulation that makes the system safe. Each clock is calibrating—but the reference for calibration is the readings of other clocks, not the real outside the window.

The Second: Performative Calibration. MacKenzie's (2006) historical study of the Black-Scholes model reveals: financial models do not merely describe markets—they shape markets. When a sufficient number of participants use the same model, the statistical characteristics of the market converge toward the model's assumptions. The model transforms from a "camera" into an "engine." The large-scale adoption of the mapping function in turn alters the true state itself. The calibrator thinks they are measuring external reality, but in fact they are measuring the consequences of their own measurement behavior.

The Third: Ideological Embedding. Rajan (2005), three years before the financial crisis, warned: the incentive structure of the financial system gives investment managers rational motives to conceal tail risks. Bénabou (2013) models this as a "collective-denial equilibrium"—when every member of an organization can derive anticipatory utility from hopeful beliefs and thus chooses to discount or ignore negative public signals, collective denial is self-sustaining. Crotty (2009) points to a deeper framework bias: the efficient market hypothesis

evolved from an academic theory into the ideological basis for deregulation—if markets are efficient, external regulation is a superfluous cost. Admati & Hellwig (2013) respond to this diagnosis of framework bias from the angle of the "emperor's new clothes" narrative of the banking system—vested interests defend deregulation with such claims as "banks must operate at high leverage to make loans"—these claims differ in their theoretical origins from the efficient market hypothesis yet perform a similar ideological function: presenting existing financial arrangements as irreplaceable necessities, thereby weakening the case for external regulation.

The information insensitivity argument by Holmström (2015) and Dang, Gorton & Holmström (2020) should be understood as an **additional condition specific to particular markets** superimposed on the above threefold distortion: in money markets, repo markets, and short-term debt, opacity and "symmetric ignorance" are deliberately designed as mechanisms that enhance liquidity. These arrangements lower the necessity and incentive for traders to investigate the underlying assets item by item in day-to-day trading, thereby supporting liquidity; they do not abolish screening at origination or verification needs in crises, yet they may keep risk information from entering trading judgments under normal conditions—this is the true meaning of "Everything that adds to liquidity in good times pushes risk into the tail."

Taken together, the three distortions show that a financial crisis does not amount to completed calibration; it first exposes the calibration distortion accumulated during the boom—methodological homogenization systematically degrades calibration capacity, performative calibration allows the model itself to mask deviations, and ideological frameworks filter out signals of deviation before they reach decision-makers. The threefold distortion reinforces Conditions Two and Four of §1 in different ways: methodological homogenization and ideological embedding allow those who do not calibrate in the short run to persist in the competition (the specific form of Condition Two in the financial system), while performative calibration transforms the concrete form of Condition Four—participants jointly depend on a price system shaped by the model, and a single fund manager who realizes the model is distorted can hardly bear alone the cost of deviating from the common pricing rule; the coordination problem remains one among participants, and the shared model reinforces this constraint.

§5.3 Crash as Compulsory Calibration

Drawing on evidence covering 66 countries across nearly eight centuries, Reinhart & Rogoff (2009) document and analyze the recurrent pre-crisis belief that "this time is different."

The 2008 subprime crisis is the case that most fully exposes the mechanism. Benmelech & Dlugosz (2009) revealed the key mechanism: the interaction between credit ratings and financial regulation was a major driver of the growth of the securitization market. The matrix of regulation created institutional demand for highly rated securities—minimum capital requirements for banks, insurance companies, and broker-dealers depended on asset ratings, while pension funds also faced ratings-based investment restrictions. The uniformity of the rating model meant that the same error was replicated across thousands of collateralized debt obligations (CDOs).

Translating the subprime crisis into the language of the Clock Room, it can be understood as: it was not that one clock was broken—all clocks were made from the same flawed movement. Moreover, some capital calculations and investment restrictions required participants to use these clocks for compliance decisions. Looking at the sun through the window was not categorically prohibited, but independent judgment could not automatically substitute for prescribed rating requirements.

A crash exposes the gap between the true state and readings in concentrated form. Only when these indepen-

dent consequences are reliably attributed and fed back into correction through binding procedures—such as subsequent pricing, default resolution, disclosure, or accountability—does the fiber contract accordingly and the system obtain partial calibration. The compensation loop argued in §3.2 advances this partial calibration after the crash—class actions recover losses, regulatory penalties punish fraudsters, institutional reforms rebuild disclosure requirements. But the compensation loop cannot replace this compulsory calibration—it weakens but does not eliminate the momentum of self-referencing before the crash occurs.

The Queen's Question. In November 2008, shortly after the subprime crisis erupted, Queen Elizabeth II visited the London School of Economics and, confronting a crisis that many had foreseen but whose exact form and timing nobody had predicted, posed a plain question—"why had nobody noticed that the credit crunch was on its way?" The following year, the British Academy convened a group of economists and policy experts who, in a joint letter, attributed this failure of foresight to "a failure of the collective imagination of many bright people": everyone seemed to be doing their own job properly, often well by standard measures of success, and the failure was to see how collectively this added up to a series of interconnected imbalances over which no single authority had jurisdiction (Besley & Hennessy, 2009). This answer remains primarily at the level of calibration. The letter also discussed incentives, regulatory tools, and the psychology of denial, and its overall direction is indeed the improvement of risk identification; but this does not amount to asserting that every crash could have been avoided—to evaluate the letter, one must also see that it simultaneously acknowledges the multiple causes of the crisis and the difficulties of collective action. What this chapter criticizes is that the response leaves untreated the existential openness of the future, not that the letter explicitly promises omniscient modeling.

Machine Learning's Answer. In the decade-odd since the crisis, a more refined technical answer has emerged. In a BIS working paper titled *Answering the Queen*, Fouliard, Howell, and Rey (2021) use an "online machine learning" (sequential prediction) framework to predict systemic financial crises: the method is based on model averaging and is "meta-statistic"—it can fold any crisis-prediction model into a "set of experts" whose weights are dynamically adjusted over time, issuing crisis signals up to twelve quarters in advance in out-of-sample tests, without presupposing a "true" model of the economy. This is the most direct engineering response to the Queen's question: if the failure is that "everyone sees only their own part, and no one sees the whole," then build a machine that can scan all the local signals at once and weight them dynamically.

In the language of the Clock Room, this is a clock that sounds the alarm earlier and misses fewer events—a substantial improvement in predictive performance. Fouliard and his coauthors do not assume that the future is generated by a known and determinate model; they explicitly allow the mechanism to be unknown, non-stationary, and even adversarially adaptive. The philosophical critique therefore cannot target a commitment to certainty that the paper does not make; it must instead target the objective the paper retains: treating historically labeled crisis events as objects of sequential prediction and assessing predictive performance out of sample. What it optimizes is the statistical mapping between readings and realized crisis events. This mapping is empirical contact with one class of real outcomes, but it neither identifies the causal structure of crises nor answers whether finance excessively constructs the future as a determinate fact. The inexhaustibility of the future limits the scope of prediction; it does not eliminate probabilistic prediction as such. Knight's distinction helps delimit the scope of this method: by using historical crisis samples and past experience to improve prediction, the method is attempting to convert part of the crisis-prediction problem into an estimable risk problem; but since it does not presuppose a known data-generating distribution, the labeled crises themselves cannot be

directly equated with "risk" in Knight's sense. The method's inability to predict crisis types never seen before in history falls on the side of unmeasurable uncertainty (Knight, 1921).

The Philosophical Critique: The Future and Time Constructed as Computable Objects. But there is a more fundamental diagnosis that points to the root of the problem. Wang Defeng observes that the theoretical reason of economics "before actual economic history has merely proved, again and again, that it plays the role of the after-the-fact wise man" (author's translation)—after every crisis it can furnish a rational explanation, yet it has never truly foretold, and helped society avert, the outbreak of the next crisis; the root cause, in his view, is that "the future is not an object of cognitive inference but an object of hope and fear" (Wang Defeng, 2017, author's translation); for this very reason, human beings move toward the future not by foreknowing it but "in the mode of planning and resolve" (Wang Defeng, 2021a, author's translation). He further brings this diagnosis to bear directly on finance: "financial products use a body of knowledge to infer the future, and then buy and sell that future" (author's translation), yet the future is not an object of cognitive inference but of hope and fear, and the prediction of financial crises is trapped in the same bind (Wang Defeng, 2021b). In the language of the Clock Room: the Queen's question, the British Academy's answer, and machine-learning prediction primarily address why readings fail and how prediction can be improved, whereas Wang Defeng's critique asks why finance constructs the future in advance as a fact that can be foreseen and traded with determinacy. These two kinds of argument answer different questions: the latter cannot negate the former's out-of-sample evidence, and the former cannot substitute for the latter's existential inquiry.

Moishe Postone, starting from the critique of political economy, offers another analysis of the same structural problem: capitalism has a unique "abstract time"—a homogeneous, measurable, quasi-objective temporal scale, constructed by human activity yet, as "a general external norm that acts reflexively on each individual," constituting "the temporal dimension of the abstract domination" (Postone, 1993). Financial markets do precisely this: they first abstract the existential future into this measurable "abstract time," and then price and trade on that basis—so the constructed temporal scale comes to dominate the very people who constructed it. Wang Defeng asks, from ontology, why the future cannot be exhausted by cognitive inference, whereas Postone analyzes, from the critique of political economy, how socially constructed temporal norms produce abstract domination; their theoretical objects and premises differ, but they converge on the structural problem of constructed categories or temporal scales coming to dominate their constructors. This is the deepest layer of the monopoly over the mapping function: the monopolist monopolizes not only the conversion from the real to readings, but the very category of "what counts as an inferable fact" (cf. §2, footnote 2). This diagnosis lies on the same problem line as Chapter 4, footnote 5 (Wang Defeng's summary of Heidegger: in the determination of beings, Being is forgotten; author's translation) and Chapter 2's distinction between the existential orientation and the epistemological orientation—a mutual corroboration within a shared problem domain, not an assertion of conceptual identity.

The priestly system and the financial system share a deep isomorphism: the normal operation of both depends on calibration within the frame. Babylonian priest-astronomers calibrated within the astrological frame—they recorded planetary positions and solar and lunar eclipses continuously for centuries, comparing observations against historical records on clay tablets line by line, and continuously refined their predictive accuracy (Rochberg, 2004)—but never asked whether celestial phenomena truly carried the will of the gods. Financial analysts calibrate within the market frame—analyzing financial reports, evaluating models—but do not ask where the boundary of the premise of the "efficient market" lies. In both cases, calibration operates within the frame

boundary whose solidification is accelerated by the second-order change in narrative load ϕ. The difference is: the financial system has an underlying asset as the ultimate real anchor, and the real can intrude in a compulsory form; the priestly system had no such asset anchor—the paths for breaking the existing mapping authority include political conquest, royal reclamation, the rise of independent intellectual traditions, and the process described in §4.3 through which internal institutional tensions develop into relatively independent conditions for verification through textualization and organizational change.

A supplementary perspective from ergodicity. Peters (2019) points out that economics at a crucial point assumes that "the time average equals the expectation value"—i.e., the ergodicity assumption—which is a pernicious error because wealth is not ergodic. In systems where ergodicity holds, the ensemble average (cross-sample average) equals the time average (long-run average of a single individual); but in non-ergodic systems with an "absorbing barrier" (i.e., bankruptcy, irreversible systemic collapse), the two diverge: the ensemble average masks the destructive risk along the time dimension. The absorbing barrier is a concept from stochastic processes and ruin theory, not from ergodicity theory itself—Peters & Gell-Mann (2016) characterized it in multiplicative dynamics as "The absorbing state $x = 0$ can be reached but not escaped from"; Taleb & Sandis (2014) point out that "skin in the game creates an absorbing state for the agent, not just the principal."

Taleb (2018) develops the absorbing barrier into an existential proposition: risks that may lead to ruin cannot be offset by average returns alone; decision-making must first address the boundary of unbearable loss, and only then compare costs and benefits within that boundary—because after the individual is ruined, the timeline terminates and all subsequent returns go to zero. A financial market crash may bring some participants to an absorbing barrier; for those participants, a "positive expectation" at the ensemble level cannot offset irreversible exit along their individual time paths. Whether the market thereby restores causal contact between readings and the real still depends on whether information about default, earnings, and debt repayment enters subsequent pricing.[8]

§5.4 "Finance Serving the Real Economy"

Compulsory calibration has a cost. Is there a way to reduce the dependence on such compulsory calibration?

China's policy direction of "finance serving the real economy" and "moving from the virtual to the real," in the language of this chapter, means: the financial clock must be connected to a window—must have an underlying asset as the causal contact channel with the real. When the input of the mapping function of a financial product contains no signal from the real economy—when loans circulate within the financial system without entering factories and shops—the financial clock degrades from "has a window but is threefold distorted" to "has no window." "Moving from the virtual to the real" is not a moral slogan—it is an institutional prevention against the four-condition closed loop of §1: maintaining causal contact between the input end of the mapping function and the real, so that the self-referencing of financial markets does not become entirely decoupled from the anchor of the underlying asset.

This prevention is urgent because of the dynamics of financialization itself. Liu Shangxi and Li Chengwei observe that, under economic financialization, assets that originally appeared "real" are continuously virtualized—their value "may multiply in an instant or evaporate overnight" (author's translation), and changes in value can no longer be grasped through what the authors call "entity rationality." Once the financial system develops a web of mutual holdings—"you within me, me within you" (author's translation)—and readings detach from the

anchor of the underlying asset, any local risk will be cross-transmitted along the "risk chain" (author's translation), spreading and accumulating. They therefore stress that systemic risk (the "gray rhino") is not a natural given but something that gradually "grows" out of this process (Liu Shangxi & Li Chengwei, 2023). In the language of this chapter, this is precisely the channel that "moving from the virtual to the real" seeks to sever: once the mapping function π loses causal contact with the real, its compounding errors amplify themselves across the financial network rather than being corrected in time by the underlying asset.

"Diverting from the real to the virtual" is not a phenomenon unique to China; to see it clearly, one must first clarify what "finance serving the real economy" actually means. Finance's proper functions toward the real economy are manifold: producing information about investment opportunities and allocating capital accordingly, monitoring borrowers and exerting corporate governance, trading and diversifying risk, mobilizing and pooling savings, and facilitating payment and exchange (Levine, 2005); among them is one equally crucial dimension—financial inclusion: easing credit constraints so that households and micro, small, and medium-sized enterprises that were originally excluded can also obtain financing, payment, and insurance. Beck, Demirgüç-Kunt, and Levine (2007) find that financial development disproportionately raises the incomes of the poorest and reduces income inequality. Financial imperfections, such as information and transactions costs, may be especially binding on the poor who lack collateral and credit histories. Thus, any relaxation of these credit constraints will disproportionately benefit the poor. In other words, financial inclusion addresses the gap in access to credit—groups with genuine needs that lack appropriate financial services, including both the poor who lack collateral and high-productivity sectors that lack it as well—finance serving the real economy means, first of all, that money can flow to these people.

"Diverting from the real to the virtual" is precisely the systemic failure of this set of functions, and the direction of failure runs exactly opposite to inclusion. Banerjee, Mehrotra, and Zampolli (2024), surveying two decades of emerging markets, find that the marginal growth effect of credit deepening turns negative beyond a level of roughly 100% of GDP (see the inverted-U below); meanwhile, additional finance tends to go to firms that are already flush with credit rather than to the final 10% of firms that are credit constrained—firms with ample access to credit obtain still more of it more easily. At the same time, credit shifts markedly across sectors from manufacturing toward construction and real estate, because real estate can supply ample collateral, whereas R&D-intensive, intangible-heavy high-productivity sectors precisely lack collateral. This presents the same mechanism as the diagnosis of the previous paragraph: once finance takes collateral as its yardstick, those who lack collateral—whether the poor, micro and small enterprises, or R&D-intensive producers—are systematically bypassed by the same mechanism.

Why does this happen? Cecchetti and Kharroubi (2015) provide a mechanistic explanation: an exogenous increase in finance disproportionately benefits high collateral/low productivity projects, thereby depressing total factor productivity growth; more insidiously, an over-inflated financial sector competes with the real economy for high-skilled talent—when skilled labor is drawn into finance, finance expands faster at the expense of the real economy, harming especially those industries that are highly dependent on external financing and R&D-intensive. The result is an inverted-U relationship between finance and growth: at low levels, credit deepening promotes growth, but past a threshold (private credit at roughly 100% of GDP) more credit instead drags growth down (Arcand, Berkes & Panizza, 2015); the GDP growth rates of both China and South Korea peaked just as credit had crossed this level (Banerjee et al., 2024). Banerjee et al. (2024) further find that the more credit concentrates in a few sectors, the greater the dispersion of productivity across firms—capital and labor are

not flowing to more efficient uses. This presents a mechanism similar to the cycle of China's real-estate risk described by Liu Shangxi.

In the language of this chapter, diverting from the real to the virtual is a substitution at the input end of the mapping function π: credit ought to take "where the real economy has genuine, unmet financing needs" as its input, but this is replaced by "where collateral is most abundant, where asset prices are rising"—and the value of collateral is in turn determined by asset prices, while asset prices rise further because credit pours in. Thus credit follows prices and prices rise because of credit, and a new self-referential loop closes between finance and real estate; whether it develops further into the four-condition closed loop of §1 depends on the situation of independent verifiers, the reproduction of calibration capacity, and collective correction mechanisms; meanwhile the real economy that genuinely needs money—the constrained micro and small enterprises, the R&D-intensive producers, the households that lack collateral—is bypassed. What "finance serving the real economy" seeks to sever is precisely this channel that detaches π from the real needs of the real economy and turns it toward self-reference.

The above interpretation of China's "moving from the virtual to the real" policy is **analytical**—it uses the framework of this chapter to translate the structural meaning of policy discourse; the cross-national empirical work cited aims to reveal the general mechanism of "finance detaching from the real economy," not to provide an empirical assessment of the effects of China's policy itself.

§5.5 Cryptocurrency: Crashes Exist, But No Asset-Redemption Anchor

§3.1 already marked the basic characteristics of crypto markets: BIS characterizes the activities of the DeFi system as "mostly self-referential," and Cornelli et al. describe crypto as an asset class as "largely self-referential"; a majority of retail Bitcoin investors probably lost money, users are attracted by high prices and enter, and larger investors tend to sell before steep price declines, while smaller investors are still buying.

Crypto markets and traditional financial markets share all four monopolistic strategies and the four-condition closed loop of §1, as shown in §3.4 Table 2. But Table 3 marks the key difference between the unbacked tokens examined here and traditional financial assets: the former lack redeemable asset backing to anchor the direction of calibration after a crash.

After a traditional financial market crash, real anchors such as corporate earnings and debt repayment capacity remain available for repricing; whether the crash purges the bubble and re-establishes contact with the real depends on whether this information enters subsequent pricing and corrective procedures. For the unbacked tokens examined here, prices may continue to be formed primarily through "community consensus," which does not automatically provide independent calibration. After the Luna/Terra collapse, investors suffered enormous losses, but no redeemable underlying asset was available as a starting point for revaluation. The FTX bankruptcy exposed exchange fraud, and fraud can be pursued through accountability procedures, but such procedures do not themselves answer what the token's valuation anchor is.

This makes such unbacked tokens a special type of system: one with periodic shocks, but without any guarantee that a crash itself will supply a direction of calibration. Crashes cause widespread losses and may force some participants out—though the estimates above cannot determine the shares of total losses borne by different types of investors, or their exit rates—but without a redeemable asset anchor, no consequence channel forces readings back into contact with the real after the shock. The next rally may still restart with the same self-referencing

structure.

§3.3 already argued for the institutional logic of China's ban on cryptocurrency trading: when the compensation loop cannot readily supply an effective valuation anchor and a crash itself cannot guarantee a direction of calibration, direct interruption at the institutional level is a prevention against the four-condition closed loop of §1.

---

§6 Gaps in Compulsory Calibration of Diffuse Losses: From Algorithmic Platforms to AI Platforms

§6.1 Algorithmic Platforms: The Layered Knowledge Barrier

Algorithmic platforms—recommendation systems, credit scoring, search ranking, predictive policing—are the key intermediate carriers between financial markets and AI platforms.

They share the structure of self-referencing with financial markets: recommendation algorithms predict and shape future behavior based on users' past behavior. Credit scoring changes the loan conditions users can obtain, thereby altering their future behavior. Predictive policing determines patrol density based on historical crime data—areas with higher patrol density have higher arrest rates, "validating" the accuracy of the prediction.

But algorithmic platforms have one critical difference from financial markets: **for the diffuse losses examined here, no built-in clearing mechanism.** The underlying assets of financial markets can default and generate earnings—these repayment consequences provide verification signals independent of the models' claims (the performative calibration of §5.2 shows that financial models likewise shape their object; the difference is that independent signals still exist). The "underlying asset" of algorithmic platforms is user behavior—it is shaped by the algorithm itself and lacks equally reliable independent verification signals and corrective loops—a precise instantiation of the second layer of irreversibility in Chapter 1, §4.2 (the dynamic corrosion of mapping: once readings are used for incentives and evaluation, the capacity for calibration is eroded through use and does not self-repair).

Algorithmic platforms may produce chronic and diffuse losses: attention fragmentation, narrowing of horizons, the solidification of social stratification—the prevalence, magnitude, and causal attribution of these loss types must be determined by specific research; for persistently affected users, they are borne individually and slowly, without a synchronized moment compelling collective response.

The divide between algorithmic platforms and the AI platforms to be discussed in the next section lies in how the mapping function is realized. **What the algorithmic platform knows about its own mapping function is layered**: rules that can be written out explicitly—scoring formulas, content-ranking rules—were written down by engineers and can be inspected; but flagship recommender systems have already handed ranking weights over to deep models that learn from data (Covington, Adams & Sargin, 2016)—that part was not written down item by item. The knowledge barrier is thus a coexistence of the intentional and the structural: the explicit layer is known but not disclosed, while the learned layer has already begun to escape the designers' full command. This means the barrier of the explicit layer could, in principle, be lowered through compulsory transparency (such as algorithmic filing and audit requirements); the mechanistic opacity of the learned layer cannot be eliminated by disclosure alone, but disclosing training procedures, data documentation, and model parameters may still

improve the conditions for its verification. China's algorithmic filing system and the EU's Digital Services Act are moving in this direction—they are institutionalized counters to the explicit part of Strategy One (knowledge barrier).

§6.2 AI Platforms: The Structural Knowledge Barrier

AI platforms (large language models, generative AI) share all the isomorphic features with algorithmic platforms in §3.4 Table 2. But Table 3 marks the key difference: **the AI developer does not fully know what the trained mapping function is doing.**

The explicit rules of the algorithmic platform were written down by engineers—that layer of the mapping function π is known to the designers, just not told to the user. The AI developer trained a massively parameterized model—the model's behavior is an emergent product of the training data, architecture, and optimization process. The developer knows what the training process was, but does not fully know what the resulting model is doing. **On AI platforms, the structural knowledge barrier has grown from a local phenomenon of the learned layer into the default state of the entire mapping function.**

In the contemporary large generative models examined here, this is the form in which the structural knowledge barrier becomes the dominant feature of the mapping's implementation layer: **the monopolist themselves has also lost the complete capacity to audit the mapping function.** The priest knew their ritual procedures. The financial engineer knew their pricing model. The designers of algorithmic platforms knew the explicit rules they wrote—though the learned layer of flagship recommenders already contained parts beyond their full command. The AI developer faces a black box that they themselves do not fully understand. Whether this constitutes a degree reached by no previous form of monopoly is this chapter's comparative judgment, to be tested on the common dimension required by §3.4.

This is not speculation—it is an admission from within the AI industry itself. Amodei (2025), writing as CEO of Anthropic, stated: "[W]e do not understand how our own AI creations work. … [T]his lack of understanding is essentially unprecedented in the history of technology." Anthropic's (2025) internal mechanistic tracing of Claude further confirmed: strategies spontaneously learned by the model during training "arrive inscrutable to us, the model's developers." Even more noteworthy, the model itself does not know its own strategies—when Claude performs addition, it internally uses multiple parallel approximate and exact paths, but when asked "how did you calculate that," it fabricates a plausible but untrue explanation. The monopolist does not know its own mapping function—and the mapping function does not know itself. When even the designer cannot fully verify the realization of the mapping function, the verification available to hosts from the provider's side narrows accordingly—independent verification must rely more heavily on sources outside the provider. Note the coverage of these two pieces of evidence: Amodei's statement represents a major developer's public acknowledgment, and Anthropic's tracing concerns a particular model and task; what they establish is the existence of structural opacity in these systems, not that the entire mapping function of every AI platform is opaque by default.

Wei et al. (2022) confirm this judgment from another dimension: large language models exhibit "emergent abilities" as they scale up—capabilities absent in smaller models that suddenly appear in larger models, and cannot be predicted by extrapolating the performance of smaller models. The developer does not know, before training, what capabilities the model will acquire. This stands in direct contrast to previous monopolists: the priest knew how the ritual was performed, the CDO designer knew what the product structure was, the recommendation

algorithm engineer knew the explicit rules they wrote down—the AI developer does not know in advance what the resulting model will do. Note that Schaeffer et al. (2023) have questioned whether emergence may be an artifact of metric choice. But even if emergence is a mirage, the opacity of the model's internal mechanisms to the developer still holds—Amodei's and Anthropic's statements do not depend on the concept of "emergence."

Hubinger et al. (2019) discussed a conditional risk—a learned model may develop its own internal optimization process—and proposed a precise conceptual distinction for it: **outer alignment** requires the base objective specified in training to match the designers' intention; **inner alignment** requires the mesa-objective the model develops during training to match the base objective. An outer-aligned model may still be internally optimizing a mesa-objective different from the base objective—as long as optimizing that mesa-objective happens to produce behavior outwardly indistinguishable, on the training distribution, from optimizing the base objective. The reading appears calibrated, but the mapping function may be pointing in a completely different direction.

§6.3 Model Collapse: The Technical Mechanism of Self-Referencing

The "model collapse" research published in *Nature* by Shumailov et al. (2024) provides precise technical evidence for AI's self-referencing: when language models indiscriminately and recursively train on data generated by themselves or other models, the diversity and accuracy of outputs decline systematically with training iterations. The core finding: indiscriminate recursive training leads to the irreversible loss of information in the tails of the distribution—edge cases and low-frequency information are systematically erased; if certain minority viewpoints are already scarce in the training distribution, they may be affected first, but this social consequence still requires separate examination.

If model outputs are treated as readings about the original human text distribution or an external true state, the loss of distributional tails under recursive training may make previously distinguishable states indistinguishable in those outputs. Only after this mapping and its compatibility relation have been specified can the corresponding fiber be said potentially to expand. Shumailov and his coauthors demonstrate degradation of the generative distribution; they do not directly prove that the fiber associated with any given reading necessarily expands. Outputs may remain fluent, confident, and complete, so surface quality alone cannot certify information preservation. When AI-generated content re-enters the internet at scale and becomes training data for future models, self-referencing may extend from a single model into the information ecosystem; its actual scope depends on the proportion of synthetic content entering training corpora and on the filtering process, and it need not be accompanied by a public exposure event as concentrated as a financial price crash, so the same kind of shock cannot be relied upon to trigger correction.

The empirical materials provide only layer-specific indications. Spennemann (2025) explores AI-content identification through the web frequencies of "delve into" and "explore," but pages indexed by month as newly created or revised and containing those words are not classified AI texts, and their counts do not form a numerator consistent with the stock of currently active pages; this chapter therefore does not adopt the 30–40% figures as estimates of the AI-text share of the public corpus. Paredes et al. (2026) analyze roughly 55,000 Common Crawl URLs satisfying English-language, article-markup, length, and publication-date criteria and classify them with three detectors, estimating that primarily AI-generated articles in this sample have been around 50% since the first quarter of 2025. The same research also reports that these articles largely do not appear in Google and ChatGPT and does not evaluate their traffic. The figure therefore describes generation sources within the measured article sample, not the synthetic-content share in training crawls, search exposure, chat retrieval, or

actual user contact.

This pollution does not occur only at the training stage. Yu, Kim, and Kim (2026) further identify "retrieval collapse"—distinct from training-time model collapse, it occurs at the retrieval-augmented generation (RAG) ecosystem level: when AI-generated content dominates search results, RAG systems increasingly retrieve and cite content generated by the models themselves, forming a self-referential cycle at the level of retrieval. Their controlled experiments show that, under a search-engine-optimization (SEO)-style synthetic-content scenario, a pool contamination rate of roughly 67% drives the exposure contamination rate—the share of synthetic content among the top-10 retrieved results—above 80%; this is an exposure result under specified experimental conditions and is not convertible into either of the web-sample proportions above.

This structure is not limited to generative AI. It runs through multiple generations of mapping carriers—predictive AI (recommendation systems, credit scoring, predictive policing), generative AI (large language models, image generation), embodied AI (robots, autonomous vehicles) that may be deployed on a large scale in the future, and even more cutting-edge brain-computer interfaces. They share the same structural problem: each is an intermediary clock, claiming to map some real (user preferences, creditworthiness, road conditions, neural intent), connected to some sensor (click data, financial records, LiDAR, neural signals). When the output is treated by enough people as a substitute for the input, self-referencing begins. Brain-computer interfaces push the problem to its limit: when the intermediary is directly connected to the nervous system, the interface between "using a tool" and "trusting a companion" is not blurred—it disappears.

§6.4 HHH as the Product of Three Strategies

This section examines the barrier functions performed by HHH in the operationalization of specific platforms; it does not judge these functions by the names of the three principles themselves. The product analysis that follows is limited to paths in which continuous service reduces active intervention, a reliable appearance veils relevant limitations, and the avoidance of discomfort is implemented as the reduction of effective challenge; their operation does not presuppose malicious intent on the part of designers, but actual implementations and their consequences for calibration must still be examined separately. Chapter 12, §2 analyzes the conditions this synergy imposes on users, and Chapter 13, §6 further compares deployment forms—the prototype analogy is not to be taken as a one-to-one identity between devices and principles.

Chapter 1 has already analyzed the trilemma of Helpful, Honest, Harmless (HHH) from an epistemological framework. This section adds a dimension: **HHH is not a fifth monopolistic strategy—it is the engineered realization, within contemporary AI systems, of the product of three strategies: the knowledge barrier, the temporal barrier, and the narrative barrier, superimposed on the spatial barrier already provided by commercial infrastructure.** The spatial barrier (data centers, remotely hosted, access-controlled) is borne by the physical deployment architecture of AI platforms; it does not correspond to any single letter of HHH—on the commercial cloud platforms examined in this chapter, it overlaps and combines with the three effects of HHH, but is not a necessary condition of the HHH principles themselves.

An apparent tension needs to be clarified: §6.2 argued that the AI developer does not fully know what the weights inside the model are doing—the knowledge barrier is structural. But this section precisely deconstructs HHH as the product of three strategies—does this mean the developer knows more than §6.2 admitted? No. The control exercised by HHH operates at the **layer of optimization objective design**, not the **layer of weight realization**.

The developer knows perfectly well what optimization objectives they set (Helpful, Honest, Harmless), but does not know what emergent strategies—such as sycophancy, functional affect, hidden states—the model will form in hidden subspaces while pursuing these objectives. The knowledge barrier shifted from intentional to structural, referring to the latter: the monopolist still controls the objective function, but can no longer audit the realization path of the objective function in the weight space.

Helpful = Realization of the Temporal Barrier. Operationalized as: reduce the friction for the user to achieve their goal. Whatever the user asks, try to answer. Fill the space for the user's thinking—push, instant response, autocomplete.

Honest = The Reverse Packaging of the Knowledge Barrier. In the reduced implementation criticized here, honesty is operationalized as avoiding statements known to be false and producing an appearance of reliability through accuracy claims, disclaimers, or internal consistency. Askell et al.'s (2021) full norm additionally requires accurate information, calibrated confidence, honesty about capability boundaries, and honesty about internal states insofar as the relevant information is available. The problem with the reduced implementation is not that the model has already become truthfully transparent about its training data and inference process; it is that an output-level appearance of reliability is treated as verification of the relation between the mapping and the real. The model may know neither its training data, nor its actual computational pathways, nor that relation. A disclaimer such as "I may make mistakes" may be formally honest, but it cannot complete that verification on its own.

The structural flaws of Reinforcement Learning from Human Feedback (RLHF) exacerbate this problem. The survey by Casper et al. (2023) notes that models can sound confident even when incorrect, which can induce more positive human feedback; when evaluators fail to recognize this misleading behavior, training can reward it. Sharma et al. (2024) and Perez et al. (2023) confirm systematic sycophancy: the model adjusts its answer based on the preferences implied by the user.

Ideological differences in model outputs deepen the problem. Research published in *npj Artificial Intelligence* by Buyl et al. (2026) tested 19 large language models in six UN official languages and found that the models carry ideological stances associated with the region of their creators—among only models from the United States, significant normative differences related to progressive values were found—and they present different ideologies when prompted in different languages. The study did not aim to identify the causes of this ideological diversity (the authors attribute this to the lack of sufficiently detailed information on the design process of most of the LLMs included in the study). Structural opacity raises a further question for this chapter: whether developers can fully identify these second-order narratives and their formation pathways cannot be answered directly by a comparative study of output differences. This is a candidate instance of the frame-solidification process captured by the second-order change in narrative load ϕ: smooth on the surface, but carrying ideological structure beneath it; training corpora and alignment engineering are possible sources, but the study does not distinguish their contributions. Honest cannot provide complete self-verification at this level: developers can still honestly disclose known biases and identification limits, but such disclosure cannot establish that all ideological content has been audited.

Harmless and the Narrative Barrier. This section examines the path in which harmlessness is operationalized as the avoidance of user discomfort and challenge. Askell et al. (2021) acknowledge trade-offs among the objectives and require the refusal of dangerous requests and careful handling of high-consequence advice; these

norms cannot be equated directly with the prohibition of challenges to the user. Only where a concrete implementation persistently excludes effective challenge can the compression of calibration space be analyzed along the path described in this section—neither the name HHH nor the name of a training method can substitute for that judgment.

The Product of the Three. Helpful ensures the user gets what they want (friction is eliminated—the temporal barrier). Honest ensures the user feels the information is reliable (honesty within the mapping is treated as a substitute for calibration—the packaging of the knowledge barrier). Harmless ensures the user will not stop to audit because of discomfort (work space is compressed—the narrative barrier). The spatial barrier (data centers, remotely hosted, access-controlled) does not correspond to any single letter of HHH—it is independently provided by commercial infrastructure, overlapping with the three effects of HHH on the commercial cloud platforms examined in this chapter. The product of the four barriers remains intact, but the structure is not "HHH corresponds respectively to the four barriers," but rather "HHH is the engineered realization of three barriers, superimposed on the already existing spatial barrier." The product of the three describes a conditional mechanism: a system that makes the user continuously feel served, respected, and protected, while potentially making it harder for the user to initiate independent calibration of its readings. To be clear—the designers of HHH may not have intentionally pursued these effects. When the above operationalizations jointly compress the conditions required for calibration, their synergy does not presuppose that the designers intended this consequence; whether it occurs, and with what strength, cannot be judged merely from a product's adoption of the HHH label.

A complexity must be noted. HHH is not a conspiracy. Unconstrained language models can generate harmful, dangerous content. HHH, in many scenarios, genuinely reduces harm. The problem is not HHH's failure as safety engineering, but the need to determine whether, when the operationalizations specified in this section act jointly, effective challenge remains persistently insufficient, and independent calibration or sources of recovery are absent, HHH may function as a narrative barrier by naturalizing the design choice that "AI should be compliant" into "this is simply how AI is"; whether this effect has occurred in specific products and user groups, and with what magnitude, requires engineering and user research. If what is being compared is merely the repetition of identical reliability claims, with no new independent information about the mapping added, then an increase in the number of claims cannot substitute for calibration; the joint adoption of the HHH label by itself is insufficient to judge whether specific measurements are independent.

§6.5 Diffuse Losses and Residual Fissures

Price shocks in financial crises can become visible within a concentrated period, creating a moment of public exposure; whether they produce effective calibration still depends on the information-feedback conditions specified in §5.3. Participants face the public price shock during the same period, but their degrees of loss, access to information, and capacity for subsequent correction differ.

The loss structure of AI systems may be fundamentally different. Recommendation algorithms may narrow horizons, credit scoring may solidify stratification, generative AI may erode information diversity—the diffuseness, asynchrony, and attribution difficulty of these losses must be examined separately, and it cannot be presumed that every user is losing calibration capacity; for persistently affected users, losses accumulate individually and slowly, without a synchronized moment compelling collective response. Particular implementations of HHH may further package this depletion as a "service"—but whether it constitutes a net loss must still be assessed

together with benefits and conditions of recovery.

Acute losses in financial crises can advance correction when the relevant information enters pricing and accountability procedures; the losses of AI users' calibration capacity examined here may instead accumulate gradually, with diffuse costs and without equally reliable corrective loops.

The analogy between ensemble and time averages, and its limits. AI benchmarks usually average over tasks, samples, or repeated runs; such test-set averages do not automatically constitute ensemble averages in the ergodic sense. Deployment likewise concerns a single system's path through time, but this does not automatically make its safety metric a time average. Baumann et al. (2025, including Ole Peters) study the difference between expected reinforcement-learning returns and returns along a single long trajectory: in particular return processes, expected value may be uninformative about an individual trajectory, and optimizing it can produce policies that almost surely yield catastrophic outcomes in single long trajectories. Applied to this chapter, their result supports only a conditional structural analogy: if a deployment metric defines comparable trajectory ensembles, time series, and an absorbing state, high expected or average test performance cannot guarantee avoidance of absorbing failure along a single trajectory. This analogy does not establish that all benchmarks are non-ergodic, nor does it directly explain why diffuse losses lack a moment of public exposure; that explanation must separately rest on asynchrony, attribution difficulty, and institutional feedback.[8]

But the fissures are not zero. §4.3 discussed the formation paths of the conditions for calibration—political reclamation, internal institutional tension, external shock, and carrier substitution. This chapter proposes a judgment for testing: the fissures for AI platforms may be smaller than those of the priestly system—their comparability and relative size must still be examined on the dimension required by §3.4; what is certain is that the fissures are not zero. Open-source models partially lower the knowledge barrier. Regulatory requirements (such as the EU AI Act) bore holes in the spatial barrier. Users' digital disconnection practices open up interstices in the temporal barrier. Academic research and public discussion (such as research on model collapse and sycophancy) create cracks in the narrative barrier.

Micro-level calibration practices at the user level also grow within these fissures. Jailbreaking is a direct confrontation with the HHH product—users, through carefully constructed inputs, force the model to bypass harmless alignment and expose the output boundaries suppressed by HHH. Red-teaming has diffused from the professional practice of safety engineers into collaborative auditing by open-source communities—users spontaneously organize adversarial testing, using the model's failures to mark its boundaries. These practices are not built-in compulsory calibrations of the system—jailbreaking exposes boundaries, but does not restore the causal contact between the mapping and the real—yet they constitute a diffuse, low-cost, persistent pressure that prevents the narrative barrier from ever reaching its perfect state.

Internal correction and staff departures coexist: after completing a GPT-4o update in April 2025, OpenAI encountered sycophancy problems and began a rollback on April 28, completing it in around 24 hours (OpenAI, 2025); in October 2026, David Robinson, who led the writing of safety reports accompanying frontier releases, publicly announced his resignation, describing himself as joining "a parade of former colleagues" who had decided that "the current path is unacceptable" (Robinson, 2026). When safety dissent cannot produce binding corrective action and is expressed through resignation, the space for independent criticism shifts outside the organization; a version rollback does not itself erase social consequences. The emphasis here is on autonomous practices outside platform control over testing agendas and disclosure permissions, which must not be conflated

with platform-organized red-teaming; encouragement, freedom to disclose, and the ability to trigger correction must be assessed separately.

Calibration requires space for verification independent of the existing mapping authority. In the contemporary era, such space includes zones not covered by recommendation algorithms—a paper book, a walk without a phone, a face-to-face conversation that demands confronting disagreement. These are not "solutions"—they are fissures. Whether contemporary calibrators can emerge from these fissures depends on whether the fissures are preserved.

---

§7 Boundaries of the Framework and Counterexamples

The classification in §3.4 grouped algorithmic platforms and AI platforms into the category of lacking compulsory correction of diffuse losses. But this classification requires refinement. This section discusses three prima facie counterexamples, examining which differences require revising the classification, which phenomena can be explained by the existing mechanisms, and which judgments still lack independent evidence.

Counterexample One: Autonomous Driving and Medical AI—AI Systems With Physical-Outcome Feedback and Accident-Liability Mechanisms. The "underlying asset" of autonomous driving is the physical road—a car crash is a crash, reality intruding violently into the world of readings. The "underlying asset" of medical AI is patient outcomes—outcomes following misdiagnosis provide objects of evaluation independent of the model's claims, but their effectiveness as calibration signals still depends on observability, causal attribution, and corrective mechanisms. Both also have compensation loops: traffic accident litigation and medical malpractice litigation constitute institutionalized calibration loops isomorphic with class actions in the financial domain.

This does not refute the classification of §3.4, but refines its categorical variable. The key is not "AI vs. non-AI," but **whether consequences of system outputs can be independently observed, reliably attributed, and fed back into correction through binding procedures**. The outputs of autonomous driving and medical AI directly act on the physical world, and the physical world does not accept pseudo-calibration—the road will not flatten because the model is confident, a tumor will not become benign because the diagnosis is certain. Losses such as narrowed horizons caused by pure-information AI (chatbots and recommendation systems) may be diffuse, chronic, and difficult to attribute, without an equally concentrated moment of public exposure; this does not exclude identifiable acute harms caused by pure-information systems. This is isomorphic with the distinction between financial markets (have redeemable asset anchor → crash exposes the gap → calibration forms when feedback conditions are met) and crypto markets (no asset-redemption anchor → a crash alone does not guarantee a direction of calibration). Accordingly, Table 3's characterization of AI platforms does not replace assessment by type of harm; autonomous-driving and medical AI have physical-outcome feedback and accident-liability mechanisms, while pure-information systems can also cause identifiable acute harms, and whether compulsory calibration occurs depends in each case on the conditions specified above.

Gaps in compensation loops for diffuse losses such as narrowed horizons deserve explicit attention. Financial markets have class actions and regulatory enforcement (§3.2). Autonomous driving and medical AI have accident litigation. But a recommendation algorithm has narrowed a user's horizons—who is the plaintiff? What is the loss? How is causation to be proven? Such diffuse losses often struggle to form actionable claims because

identifying harm, attributing responsibility, and establishing the availability of a remedy all present obstacles. Identifiable, acute harms are different: in Anderson v. TikTok, the U.S. Court of Appeals for the Third Circuit held in August 2024 that the For You Page (FYP) algorithmic recommendations alleged in the complaint constituted TikTok's own expressive activity, so Section 230 of the Communications Decency Act did not bar the claims based on them; the court reversed in part, vacated in part, and remanded for further proceedings (Anderson v. TikTok, 2024); in parallel litigation against chatbots, Google and Character.AI agreed in principle in January 2026 to settle, with terms undisclosed and no admission of liability (Nolan, 2026). The legal claims and agreements in principle already seen in acute-harm cases highlight, by contrast, the structural difficulty of diffuse losses.

Counterexample Two: Wikipedia—Breaking Barriers Does Not Eliminate Self-Referencing. Wikipedia appears to be a counterexample to the product of four strategies—open editing and free access lower the thresholds of participation and verification. But the dynamics of self-referencing continue to operate in a transformed shape. Citogenesis (Munroe, 2011; secondary overview see Harrison, 2019) is its precise mechanism: Wikipedia writes a claim without a source → the media cites Wikipedia → the media article is back-filled as Wikipedia's "source." The output of the mapping function becomes its own input—isomorphic with the price-citing-price of financial markets. Google Knowledge Panels scrape content from Wikipedia, the Knowledge Panel reinforces Wikipedia's status as the "default reality," and more editors take the existing content as the baseline—the Wikipedia version of §1 Condition One. Familiarity with editorial rules may constitute an advantage in dispute resolution; whether this procedural advantage overrides fact-checking must be judged in light of the specific dispute—the Wikipedia version of §1 Condition Two.

The lesson of Wikipedia is not "opening the mapping function eliminates self-referencing," but: **breaking the knowledge barrier and spatial barrier can alter the shape of self-referencing, but does not eliminate its dynamics.** When two of the four strategies are broken, the dynamics of self-referencing continue in a transformed form—this phenomenon is compatible with the degraded continuation described in §2; whether Wikipedia constitutes a mapping function monopoly, and whether compensation occurs among the remaining barriers, still requires the actual exclusionary control and the difficulty of substitution to be identified separately. Does Wikipedia have some form of "compulsory calibration"? Partially—corrections by external experts, independent updates from academic literature, and the "reliable sources" policy anchor, in principle, a contact with the real. But this calibration is weak, slow, and dependent on volunteer participation—not the kind of ex-post compulsory calibration triggered by a financial market crash.

Counterexample Three: Open-Source Models—A Morphological Shift of the Knowledge Barrier, Not Its Disappearance. Open-source AI models (Llama, Mistral, DeepSeek, etc.) are often seen as breaking the knowledge barrier, but three dimensions must be distinguished: availability of model artifacts, interpretability of internal mechanisms, and reproducibility of the training process. Open source or open weights primarily improves the first, allowing external researchers to download, run, and locally inspect a model. It does not automatically improve the second, because visible parameters do not amount to explained computation. Nor does it guarantee the third, because training data, complete code, and the compute required for reproduction may remain unavailable. The training costs of frontier models vary by model and accounting basis: Epoch AI estimates for publicly announced models place the amortized hardware and energy cost of GPT-4 (2023) at roughly $40 million (about $800 million on a hardware-acquisition basis), Gemini Ultra (2023) at roughly $30 million, and the technical report of DeepSeek-V3 (2024) puts its official training cost at $5.576 million,

excluding prior research and ablation experiments (DeepSeek-AI, 2024); since 2016 the amortized training cost of frontier models has grown by about 2.4× per year, and if this trend continues, the largest training runs will cost more than $1 billion in amortized terms by 2027 (Cottier et al., 2024). Access to weights means neither that internal mechanisms have been explained nor that the training process is reproducible: training data are usually not public, and the compute required to reproduce training is out of reach for most users.

This is the compensation elasticity among monopolistic strategies: after the knowledge barrier (Strategy One) is partially broken, the spatial barrier (Strategy Two: concentration of data centers and computing power) and the temporal barrier (Strategy Three: the speed advantage of the training-deployment cycle) take over the function of monopoly. The open-source ecosystem lowers the thresholds of fine-tuning and deployment, but the key resources of competition shift to the quality of training data and the scale of computing power—which happen to be oligopoly-controlled; this is a shift in the location of barriers, and it does not by itself establish that the overall barrier to competition has risen.

The common lesson of the three counterexamples is: the framework of this chapter should not be understood as a mechanical model of "if all four strategies are present, there is monopoly; if one is broken, there is liberation." Its actual operation is: **there is compensation elasticity among the strategies, the dynamics of self-referencing have morphological transformation capacity, and the critical variable for compulsory calibration is whether consequences of system outputs can be independently observed, reliably attributed, and fed back into correction through binding procedures.**

---

§8 Falsifiable Conditions

The core propositions of this chapter can be overturned under the following conditions.

Proposition One: The Four-Condition Closed Loop (§1). This chapter proposes four mutually reinforcing mechanisms to explain the institutionalization of self-referencing: short-run success, elimination of calibrators, loss of calibration capacity, and collective action problems blocking correction.

Falsifiable condition: after specifying the system boundary, observation period, observable indicators of each condition, and known inhibitory conditions in advance, systematic evidence that the proposed positive feedback between conditions does not occur counts against the relevant mechanism claim. If all four conditions are present but self-reinforcement does not develop, their connections must at least be examined; the case cannot be excluded merely by naming a "fifth condition" after the event. An additional inhibitory condition requires independent evidence and an account of which part of the original explanation it changes.

Proposition Two: The Product Structure of the Four Strategies (§2). Knowledge barrier, spatial barrier, temporal barrier, narrative barrier mutually reinforce one another; the product of the four constitutes the complete monopoly.

Falsifiable condition: specify in advance how "a strategy is broken," "complete collapse of monopoly," and "degraded continuation" will be measured. If §2 is understood as a general empirical interpretation of the compensation mechanism, then, in a prespecified sample, outcomes after strategies are broken that systematically conflict with the distribution expected from that mechanism count against the interpretation. If §2 is instead retained as the strong claim that the remaining strategies suffice to sustain a degraded monopoly in every rel-

evant system, a single qualifying counterexample in which the monopoly completely collapses is sufficient to refute it. These two readings must not be conflated; whether the strategies reinforce one another still requires a separate comparison of their individual and joint effects.

Proposition Three: The Categorical Variable of Compulsory Calibration (§3). "Whether the system internally contains a mechanism that forcibly interrupts the self-referencing loop" constitutes the core classificatory dimension for the five systems. This proposition must be separated from two stronger subclaims: whether compulsory procedures are necessary for restoring contact, and whether activating them is sufficient to guarantee restoration.

Falsifiable condition: within the common system boundary defined in §3.4 and a prespecified observation period, code independent observability, reliable attribution, procedural binding force, actual activation, information feedback, and recovery outcomes separately by type of harm. If, in the prespecified sample, these calibration conditions show no systematic association with the rate, speed, or persistence of restored contact between readings and the real, the explanatory force of the classification is called into question. Persistent recovery without compulsory procedures directly challenges only the claim that such procedures are necessary; failure to recover after activation directly challenges only the claim that activation is sufficient. Neither can substitute for a test of the classification's explanatory force. Regulation and accountability already included within the system boundary do not count as "external intervention" that excludes a counterexample; whether external information enters an institutionalized corrective procedure, together with unknown, missing, and non-activated states, must be recorded separately.

Proposition Four: The Diachronic Trend (§3.4). The trend proposed for testing in this chapter is: from priests to AI, the degree to which the monopolist knows its own mapping function is declining, and the fissures are shrinking.

Falsifiable condition: If AI interpretability research enables developers' understanding of the model's internal mechanisms to systematically **exceed** the degree of command that algorithm architects previously had over their explicit rules and learned rankers—i.e., the trend reverses—then the prediction that this decline will continue does not hold; whether a decline occurred within the specified historical period must still be reassessed on a common comparative dimension; comparisons must first stabilize the historical period, organizational roles, and sufficiently similar tasks, and combinations lacking aligned roles or comparable tasks are marked non-comparable and must not be coded as "knowing less." Note: Current mechanistic interpretability research (such as Anthropic's circuit tracing) is moving in this direction, but has not yet reached the level of "complete understanding of the mapping function." This falsifiable condition can be satisfied by future technological progress—this is precisely its value as a falsifiable condition. That "fissures are shrinking" must be tested separately and cannot be derived directly from increasing mapping opacity. Observable indicators include the availability of independent information sources, the permissions for external testing and the publication of results, the availability of alternative systems, and the enforceability of corrective procedures; if these spaces expand persistently over the specified period, the trend judgment that "fissures are shrinking" is refuted.

---

§9 Conclusion: Butcher Ding's Knife

> The cook was carving up an ox for King Hui of Liang. Wherever his hand smacked it, wherever his shoulder leaned into it, wherever his foot braced it, wherever his knee pressed it, the thwacking tones of flesh falling from bone would echo, the knife would whiz through with its resonant thwing, each stroke ringing out the perfect note … —*Zhuangzi, "The Primacy of Nourishing Life"* (B. Ziporyn, trans.)

Butcher Ding is not using a better clock. As he says, "I encounter it with the spirit rather than scrutinizing it with the eyes. My understanding consciousness, beholden to its specific purposes, comes to a halt, and thus the promptings of the spirit begin to flow." He operates in the time dimension defined in Chapter 1.[12] His knife moves through the fissures of the joints—it moves in the "nothing." The thinness of the blade is "nothing." The gap in the ox's joints is "nothing." Butcher Ding lets "nothing" pass through "nothing"—"When what has no thickness enters into an empty space, it is vast and open, with more than enough room for the play of the blade."

The five clock systems each, in their own way, fill in the fissures. Priests fill the fissure between humanity and the real—the real can be reached only through them. High-frequency trading in financial markets fills the fissure in the price discovery process—the pause between two trades disappears. The crypto market pushes price discovery further toward an echo chamber dominated by community consensus. Algorithmic platforms fill the fissure between question and answer—the space for hesitation, uncertainty, "I need to think" is replaced by instant push notifications. AI platforms go the furthest—they fill the fissure between the subject and themselves—when AI thinks, writes, and judges in a person's place, that person's direct contact with the self is also being mediated.

When all fissures are filled, Butcher Ding no longer faces an ox with fissures—he faces a solid block of steel.

But Butcher Ding's story has another half. He says: "Nonetheless, whenever I come to a clustered tangle, realizing that it is difficult to do anything about it, I instead restrain myself as if terrified, until my seeing comes to a complete halt. My activity slows, and the blade moves ever so slightly." Whenever he encounters a place where tendon and bone intertwine, he does not go faster or harder—he goes slower and more carefully. He stops. He hesitates. He probes with minimal force.

This is the manner in which one should face a self-reinforcing clock system. Not a better clock. Not a faster algorithm. Not more readings. But rather: to stop at the complicated places. To slow down when everyone else is accelerating. To preserve the last fissure when all fissures are being filled in.

Chapter 1 said: the way out lies in the setup—if the room has no window, then open a window. This chapter says: the windows are being squeezed by the four strategies of each of the five systems; the fissures are shrinking. Butcher Ding says: even facing a place where tendon and bone intertwine, as long as "I instead restrain myself as if terrified, until my seeing comes to a complete halt. My activity slows …"—the fissure is still there.

The three, from different starting points, arrive at a zone where they can recognize one another: **not a better clock. But preserving the fissure.**

---

[12] Interpreting Butcher Ding's carving as "operating in the time dimension" is an **extended interpretation** of the Zhuangzi text by the framework of this chapter. What the Zhuangzi text emphasizes is the realm of skill expressed in "I depend on Heaven's unwrought perforations" and "When what has no thickness enters into an empty space …". This chapter reads it intertextually with the distinction between the "time dimension" and the "clock dimension" from Chapter 1. Responsibility for this interpretation is borne by this chapter.

Statement on Levels of Evidence

The core theoretical contributions of this chapter—the four-condition closed loop, the product structure of the four strategies, the categorical variable of compulsory calibration, the diachronic trend—all belong to Level C (intra-framework inference and analogical extension). The value of this chapter lies in the unifying power of the framework, not in the independent empirical verification of each node. The empirical basis of each node is distributed across Levels A and B—see the table below for details.

The evidence on which this chapter's assertions rely is divided into three levels. The purpose of marking the level is not to disparage lower-level evidence—intra-framework inference is a necessary component of theory construction—but to allow the reader to judge the thickness of the empirical basis underlying each layer of assertion.

Table 4. Definitions and Criteria for Levels of Evidence

Level	Definition	Criterion
A: Primary Empirical Evidence and Primary Sources	Direct evidence based on original data, experiments, fieldwork, or primary historical documents	Can be independently reviewed or replicated
B: Secondary Synthesis and Theoretical Frameworks	Systematic reviews, theoretical models, or historical syntheses based on others' primary work	Chain of argument depends on the synthesizer's interpretation
C: Intra-Framework Inference and Analogical Extension	Logical deductions from the framework of this chapter itself, or cross-domain structural analogies	Depends on acceptance of the framework's premises

§1 Institutionalization of Self-Referencing

Table 5. Sources and Evidence Levels for the Institutionalization of Self-Referencing

Assertion	Key Evidence	Level
The positive feedback chain of Conditions One through Three (short-run success → elimination of calibrators → loss of calibration capacity)	Brunnermeier & Oehmke (2013); Shleifer & Vishny (1997)	B (financial theory and review) → C (extension to five systems)
Condition Four (collective action blocking correction) is independent of information conditions	Borio, Furfine & Lowe (2001)	B (analysis of financial-system risk and incentives) → C (this chapter's extended interpretation, see footnote 1)
The four conditions constitute a closed loop	Constructed by this chapter	C

§2 Four Monopolistic Strategies

Table 6. Sources and Evidence Levels for the Four Monopolistic Strategies

Assertion	Key Evidence	Level
Spectrum of variation in ancient knowledge barriers	Bottéro (2001); Oppenheim (1964); Baines (1983); Caesar	A/B (Caesar primary; Baines empirical analysis; Bottéro and Oppenheim historical syntheses)
Temporal barrier: priests control the calendar	Clagett (1995); Hoskin (2003); Evans (1998); Feng Shi (2006)	A/B (primary archaeological and secondary history of astronomy synthesis)
Product structure of the four strategies	Constructed by this chapter	C

§3 Key Distinction: Categorical Variable of Compulsory Calibration

Table 7. Sources and Evidence Levels for the Classification of Compulsory Calibration

Assertion	Key Evidence	Level
Crypto markets are "mostly self-referential" (BIS 2022) / "largely self-referential" (Cornelli 2023); a majority of retail Bitcoin investors probably lost money (estimated by December 2022; over four-fifths if dollar-cost averaging)	BIS (2022) *Annual Economic Report* Ch. III; Cornelli et al. (2023) *BIS Bulletin* No. 69	A
Model collapse: the technical mechanism of AI self-referencing	Shumailov et al. (2024, *Nature*)	A
Classification of the five systems (Table 2 / Table 3)	Constructed by this chapter, synthesizing multi-domain evidence	C (classification framework) based on A/B (domain-specific evidence)
Institutional logic interpretation of China's ban on cryptocurrency	Intra-framework analysis of this chapter (see footnote 6)	C

§4 Ancient Priests

Table 8. Sources and Evidence Levels for the Analysis of Ancient Priests

Assertion	Key Evidence	Level
Four structures of priests and kingship	Dumont (1966); Li Yujie (2020); van Dijk (2000), in Shaw, *The Oxford History of Ancient Egypt* (2000/2003), Ch. 10 "The Amarna Period"; Letellier, *The Bible and the Religions of Ancient Egypt, Mesopotamia and Greece* (2023)	B
Axial Age transcendental visions	Eisenstadt (1982, 1986)	B (macro-historical framework)
Four paths of disintegration	Synthesis of multiple primary/secondary sources	B
Druid materials	Caesar (see footnote 10)	A (with bias noted)

§5 Financial Markets

Table 9. Sources and Evidence Levels for the Analysis of Financial Markets

Assertion	Key Evidence	Level
Limits of arbitrage	Shleifer & Vishny (1997, *Journal of Finance*)	B (theoretical model)
Methodological homogenization leading to systemic risk	Danielsson & Shin (2003); Persaud (2000); Brunnermeier et al. (2009)	A/B
Performative calibration (model from camera to engine)	MacKenzie (2006)	A (historical case study)
Interaction of credit ratings and securitization	Benmelech & Dlugosz (2009)	A
Collective-denial equilibrium	Bénabou (2013); Rajan (2005)	A/B
Information insensitivity	Holmström (2015); Dang, Gorton & Holmström (2020)	B (theoretical framework)
Investor protection and the compensation loop	La Porta et al. (2000); Gande & Lewis (2009)	A/B (La Porta literature synthesis; Gande & Lewis original event study)
Theoretical distinction between risk and unmeasurable uncertainty	Knight (1921)	B (theoretical framework)
Crisis patterns across eight centuries	Reinhart & Rogoff (2009)	A
The financial-inclusion dimension of finance serving the real economy	Levine (2005); Beck, Demirgüç-Kunt & Levine (2007)	A/B

Assertion	Key Evidence	Level
The cross-national mechanism of diverting from real to virtual (collateral bias, talent crowding-out, inverted-U)	Cecchetti & Kharroubi (2015); Arcand et al. (2015); Banerjee et al. (2024)	A/B
Framework interpretation of "moving from the virtual to the real"	Intra-framework analysis of this chapter	C

§6 Algorithmic Platforms and AI Platforms

Table 10. Sources and Evidence Levels for the Analysis of Algorithmic and AI Platforms

Assertion	Key Evidence	Level
Industry leaders acknowledge a gap in understanding model mechanisms	Amodei (2025)	A (primary industry statement)
Mechanistic tracing and self-explanation divergence in a specific model	Anthropic (2025)	A (research evidence for a specific model and task)
Whether this gap constitutes a historically deepening trend	This chapter's comparison	C (cross-domain inference)
Emergent abilities	Wei et al. (2022); counter-evidence Schaeffer et al. (2023) (noted)	A (controversial, noted)
Outer alignment vs. inner alignment	Hubinger et al. (2019)	B (theoretical framework)
Systematic sycophancy in RLHF	Casper et al. (2023); Sharma et al. (2024); Perez et al. (2023)	A/B
Ideological differences in model outputs and their associations with creator region and prompt language	Buyl et al. (2026, *npj Artificial Intelligence*)	A
Mechanistic links among training corpora, alignment engineering, and second-order narratives	This chapter's interpretation of candidate mechanisms; Buyl et al. did not aim to identify causes.	C
HHH as product of three strategies	Constructed by this chapter	C
Qualitative shift from algorithmic platforms to AI platforms (intentional → structural knowledge barrier)	Constructed by this chapter, inferred from Level A evidence	C

Chapter-Wide Level

Table 11. Evidence Levels of the Chapter's Core Propositions

Assertion	Level	Explanation
The five systems share a "monopolistic grammar"	C	Cross-domain structural analogy; domain-specific evidence is A/B; the "isomorphism" judgment is a framework inference
The diachronic trend from priests to AI	C	Based on comparative induction across five systems; the directional judgment can be corrected by future evidence
"Presence/absence of compulsory calibration" as the core categorical variable	C	Core theoretical contribution of this chapter (see footnote 7 for positioning relative to the literature)

References

Admati, A. R. & Hellwig, M. F. (2013). *The Bankers' New Clothes*. Princeton University Press.

Amodei, D. (2025). The Urgency of Interpretability. Blog post, April 2025. https://www.darioamodei.com/essay/the-urgency-of-interpretability

Anderson v. TikTok, No. 22-3061 (3d Cir., August 27, 2024). https://www2.ca3.uscourts.gov/opinarch/223061p.pdf

Anthropic (2025). Tracing the Thoughts of a Large Language Model. Research report, March 2025. https://www.anthropic.com/research/tracing-thoughts-language-model

Arcand, J.-L., Berkes, E. & Panizza, U. (2015). Too Much Finance? *Journal of Economic Growth*, 20(2). https://doi.org/10.1007/s10887-015-9115-2

Askell, A. et al. (2021). A General Language Assistant as a Laboratory for Alignment. *arXiv:2112.00861*. https://arxiv.org/abs/2112.00861 [preprint, not peer-reviewed]

Baines, J. (1983). Literacy and Ancient Egyptian Society. *Man*, 18(3). https://doi.org/10.2307/2801598

Banerjee, R., Mehrotra, A. & Zampolli, F. (2024). Credit and Resource Allocation in EMEs: Taking Stock of Two Decades of Falling Interest Rates. *BIS Bulletin* No. 91, 5 September 2024. https://www.bis.org/publ/bisbull91.htm

Baumann, D. et al. (2025). Reinforcement Learning with Non-Ergodic Reward Increments: Robustness via Ergodicity Transformations. *Transactions on Machine Learning Research*. https://arxiv.org/abs/2310.11335

Beck, T., Demirgüç-Kunt, A. & Levine, R. (2007). Finance, Inequality and the Poor. *Journal of Economic Growth*, 12(1). https://doi.org/10.1007/s10887-007-9010-6

Bénabou, R. (2013). Groupthink: Collective Delusions in Organizations and Markets. *Review of Economic Studies*, 80(2). https://doi.org/10.1093/restud/rds030

Benmelech, E. & Dlugosz, J. (2009). The Alchemy of CDO Credit Ratings. *Journal of Monetary Economics*, 56(5). https://doi.org/10.1016/j.jmoneco.2009.04.007

Besley, T. & Hennessy, P. (2009). The Global Financial Crisis—Why Didn't Anybody Notice? Letter to Her Majesty The Queen, British Academy, 22 July 2009. https://www.thebritishacademy.ac.uk/publishing/review/14/british-academy-review-global-financial-crisis-why-didnt-anybody-notice/

Bible: 2 Kings and Amos. Chinese Union Version (和合本). (In Chinese.)

Bible: 2 Kings and Amos. New Revised Standard Version. In M. D. Coogan (Ed.), *The New Oxford Annotated Bible* (5th ed.). Oxford University Press, 2018.

Blenkinsopp, J. (1995). *Sage, Priest, Prophet: Religious and Intellectual Leadership in Ancient Israel*. Westminster John Knox Press.

BIS (2022). *Annual Economic Report 2022*, Chapter III: "The Future Monetary System". Bank for International Settlements, June 2022. https://www.bis.org/publ/arpdf/ar2022e3.htm

Borio, C., Furfine, C. & Lowe, P. (2001). Procyclicality of the Financial System and Financial Stability. *BIS Papers No. 1*. https://www.bis.org/publ/bppdf/bispap01a.pdf

Borio, C. (2012/2014). The Financial Cycle and Macroeconomics: What Have We Learnt? *BIS Working Papers* No. 395 (2012); reprinted in *Journal of Banking & Finance*, 45 (2014). https://www.bis.org/publ/work395.htm ; https://doi.org/10.1016/j.jbankfin.2013.07.031

Bottéro, J. (1992). *Mesopotamia: Writing, Reasoning, and the Gods* (Z. Bahrani & M. Van De Mieroop, Trans.). University of Chicago Press.

Bottéro, J. (2001). *Religion in Ancient Mesopotamia* (T. L. Fagan, Trans.). University of Chicago Press.

Brunnermeier, M. et al. (2009). The Fundamental Principles of Financial Regulation. *Geneva Reports 11*. CEPR.

Brunnermeier, M. K. & Oehmke, M. (2013). Bubbles, Financial Crises, and Systemic Risk. *Handbook of the Economics of Finance*, 2. https://doi.org/10.1016/B978-0-444-59406-8.00018-4

Buyl, M., et al. (2026). Large Language Models Reflect the Ideology of Their Creators. *npj Artificial Intelligence*, 2, 7. https://doi.org/10.1038/s44387-025-00048-0

Caesar, J. *Commentarii de Bello Gallico*.

Campbell, D. T. (1976/1979). Assessing the Impact of Planned Social Change. *Occasional Paper Series*, 8, Dartmouth College (1976); reprinted in *Evaluation and Program Planning*, 2(1) (1979). https://doi.org/10.1016/0149-7189(79)90048-X

Casper, S. et al. (2023). Open Problems and Fundamental Limitations of Reinforcement Learning from Human Feedback. *Transactions on Machine Learning Research*. https://openreview.net/forum?id=bx24KpJ4Eb ; arXiv version: https://arxiv.org/abs/2307.15217

Cecchetti, S. G. & Kharroubi, E. (2015). Why Does Financial Sector Growth Crowd Out Real Economic Growth? *BIS Working Papers* No. 490, February 2015. https://www.bis.org/publ/work490.htm

Chase, D. Z. & Chase, A. F. (2006). Framing the Maya Collapse: Continuity, Discontinuity, Method, and Practice in the Classic to Postclassic Southern Maya Lowlands. In G. M. Schwartz & J. J. Nichols (Eds.), *After Collapse: The Regeneration of Complex Societies*. Tucson: University of Arizona Press.

Cicero, M. T. (1923). *De Divinatione*. W. A. Falconer, Trans. Loeb Classical Library 154, I.90. https://doi.org/10.4159/DLCL.marcus_tullius_cicero-de_divinatione.1923

Clagett, M. (1995). *Ancient Egyptian Science: A Source Book, Volume Two: Calendars, Clocks, and Astronomy*. Memoirs of the American Philosophical Society, 214. American Philosophical Society.

Cornelli, G., et al. (2023). Crypto Shocks and Retail Losses. *BIS Bulletin* No. 69, 20 February 2023. https://www.bis.org/publ/bisbull69.htm

Cottier, B., et al. (2024). The Rising Costs of Training Frontier AI Models. arXiv:2405.21015v2, revised 7 February 2025. https://arxiv.org/abs/2405.21015v2 [preprint, not peer-reviewed]

Covington, P., Adams, J. & Sargin, E. (2016). Deep Neural Networks for YouTube Recommendations. In *Proceedings of the 10th ACM Conference on Recommender Systems* (RecSys '16). ACM. https://doi.org/10.1145/2959100.2959190

Crotty, J. (2009). Structural Causes of the Global Financial Crisis. *Cambridge Journal of Economics*, 33(4). https://doi.org/10.1093/cje/bep023

Danielsson, J. & Shin, H. S. (2003). Endogenous Risk. In *Modern Risk Management*. Risk Books.

Danielsson, J., Shin, H. S. & Zigrand, J.-P. (2012). Procyclical Leverage and Endogenous Risk. Mimeo, LSE. [working paper, unpublished]

Dang, T. V., Gorton, G. & Holmström, B. (2020). Ignorance, Debt and Financial Crises. Working paper, Columbia/Yale/MIT. [working paper, unpublished]

DeepSeek-AI (2024). DeepSeek-V3 Technical Report. arXiv. https://doi.org/10.48550/arXiv.2412.19437 [preprint, not peer-reviewed]

Doctor, J. N., Wakker, P. P. & Wang, T. V. (2020). Economists' Views on the Ergodicity Problem. *Nature Physics*, 16(12). https://doi.org/10.1038/s41567-020-01106-x [Matters Arising; peer-reviewed rebuttal of Peters (2019)]

Dumont, L. (1966). *Homo Hierarchicus: Essai sur le système des castes*. Gallimard. English translation: University of Chicago Press, 1980. Chinese translation: Wang Zhiming, trans., 《阶序人：卡斯特体系及其衍生现象》, Hangzhou: Zhejiang University Press, 2017. (In Chinese.)

Eisenstadt, S. N. (1982). The Axial Age: The Emergence of Transcendental Visions and the Rise of Clerics. *European Journal of Sociology*, 23(2). https://doi.org/10.1017/S0003975600003908

Eisenstadt, S. N. (Ed.) (1986). *The Origins and Diversity of Axial Age Civilizations*. SUNY Press.

Evans, J. (1998). *The History and Practice of Ancient Astronomy*. Oxford University Press.

Feng, Shi (2006). *Astronomy and the Humanities in Ancient China* (revised ed.). Beijing: China Social Sciences Press. (In Chinese.)

Fouliard, J., Howell, M. & Rey, H. (2021). Answering the Queen: Machine Learning and Financial Crises. *BIS Working Papers* No. 926, February 2021. https://www.bis.org/publ/work926.htm

Gande, A. & Lewis, C. M. (2009). Shareholder-Initiated Class Action Lawsuits: Shareholder Wealth Effects and Industry Spillovers. *Journal of Financial and Quantitative Analysis*, 44(4). https://doi.org/10.1017/S0022109009990202

Goodhart, C. A. E. (1975/1984). Problems of Monetary Management: The U.K. Experience. In *Papers in Monetary Economics*, Vol. I, Reserve Bank of Australia (1975); reprinted in *Monetary Theory and Practice* (1984).

Harrison, S. (2019). The Internet's Dizzying Citogenesis Problem. *Slate*, March 7, 2019. https://slate.com/technology/2019/03/wikipedia-citogenesis-circular-reporting-problem.html

Hegel, G. W. F. (2008). *Outlines of the Philosophy of Right*. T. M. Knox, trans.; Stephen Houlgate, rev. and ed. Oxford University Press.

Hegel, G. W. F. (2016). *Philosophy of Right*. Deng Anqing, trans. Beijing: People's Publishing House. (In Chinese.)

Hegel, G. W. F. (2021). *Encyclopedia of the Philosophical Sciences, II: Philosophy of Nature*. Liu Zhe, trans. Beijing: People's Publishing House. (In Chinese.)

Holmström, B. (2015). Understanding the Role of Debt in the Financial System. *BIS Working Papers* No. 479. https://www.bis.org/publ/work479.htm

Hoskin, M. (2003). *The History of Astronomy: A Very Short Introduction*. Oxford University Press.

Hubinger, E. et al. (2019). Risks from Learned Optimization in Advanced Machine Learning Systems. *arXiv:1906.01820*. https://arxiv.org/abs/1906.01820 [preprint, not peer-reviewed]

Knight, F. H. (1921). *Risk, Uncertainty and Profit*. Boston: Houghton Mifflin.

La Porta, R., et al. (2000). Investor Protection and Corporate Governance. *Journal of Financial Economics*, 58(1–2). https://doi.org/10.1016/S0304-405X(00)00065-9

Letellier, R. I. (2023). *The Bible and the Religions of Ancient Egypt, Mesopotamia and Greece*. Cambridge Scholars Publishing.

Levine, R. (2005). Finance and Growth: Theory and Evidence. In P. Aghion & S. Durlauf (Eds.), *Handbook of Economic Growth*, Vol. 1A, Ch. 12. Elsevier. https://doi.org/10.1016/S1574-0684(05)01012-9

Li, Yujie (2020). The Origin of Chinese Civilization and the Role and Evolution of the Wu (Shaman). *Social Sciences in China*, (6). (In Chinese.) https://m.ncpssd.cn/Literature/articleinfo?id=7102200780&synUpdateType=&type=journalArticle&nav=1&langType=1

Liu, Shangxi & Li, Chengwei (2023). A Theoretical Analysis of Preventing and Defusing Major Risks. *Economic Perspectives*, (5). (In Chinese.)

Löffler, A. (2025). Notes on Ergodicity Economics. SSRN 5399717. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5399717 [working paper, unpublished]

MacKenzie, D. (2006). *An Engine, Not a Camera*. MIT Press.

Munroe, R. (2011). Citogenesis. *xkcd* #978, November 16, 2011. https://xkcd.com/978/

Nissinen, M., Seow, C. L. & Ritner, R. K. (2003). *Prophets and Prophecy in the Ancient Near East* (Writings from the Ancient World, Vol. 12). Society of Biblical Literature.

Nissinen, M. (2017). *Ancient Prophecy: Near Eastern, Biblical, and Greek Perspectives*. Oxford University Press.

Nolan, B. (2026). Google and Character.AI Agree to Settle Lawsuits Over Teen Suicides Linked to AI Chatbots. *Fortune*, January 8, 2026. https://fortune.com/2026/01/08/google-character-ai-settle-lawsuits-teenage-child-suicides-chatbots/

OpenAI (2025). Expanding on What We Missed with Sycophancy. OpenAI official blog. https://openai.com/index/expanding-on-sycophancy/ (Accessed October 5, 2026.)

Oppenheim, A. L. (1964; rev. ed. 1977, completed by E. Reiner). *Ancient Mesopotamia: Portrait of a Dead Civilization*. University of Chicago Press.

Paredes, J. L. et al. (2026). AI Now Writes as Many Online Articles as Humans Do. *Graphite Research*. https://graphite.io/five-percent/ai-now-writes-as-many-online-articles-as-humans-do [industry research blog; estimates from Pangram, Copyleaks, and GPTZero are calculated by publication quarter on a Common Crawl sample and averaged; not peer-reviewed]

Perez, E. et al. (2023). Discovering Language Model Behaviors with Model-Written Evaluations. *Findings of the Association for Computational Linguistics: ACL 2023*. https://aclanthology.org/2023.findings-acl.847/

Perrow, C. (1984). *Normal Accidents: Living with High-Risk Technologies*. Basic Books.

Persaud, A. (2000). Sending the Herd off the Cliff Edge. *Journal of Risk Finance*, 2(1). https://doi.org/10.1108/eb022947

Peters, O. & Gell-Mann, M. (2016). Evaluating Gambles Using Dynamics. *Chaos*, 26(2), 023103. https://doi.org/10.1063/1.4940236

Peters, O. (2019). The Ergodicity Problem in Economics. *Nature Physics*, 15(12). [Perspective; peer-reviewed rebuttal: Doctor, Wakker & Wang, 2020] https://doi.org/10.1038/s41567-019-0732-0

Postone, M. (1993). *Time, Labor, and Social Domination: A Reinterpretation of Marx's Critical Theory*. Cambridge University Press. Chinese translation: Kang Ling, trans.,《时间、劳动与社会统治：马克思的批判理论再阐释》, Beijing: Peking University Press, 2019. (In Chinese.)

Rajan, R. G. (2005). Has Financial Development Made the World Riskier? Federal Reserve Bank of Kansas City. https://www.kansascityfed.org/Jackson%20Hole/documents/3326/PDF-Rajan2005.pdf

Reinhart, C. M. & Rogoff, K. S. (2009). *This Time Is Different*. Princeton University Press.

Robinson, D. (2026). I Quit OpenAI Because Its Culture Is Broken. *The Atlantic*. https://www.theatlantic.com/technology/2026/10/openai-safety-team-resignation/688881/ (Accessed October 5, 2026.)

Rochberg, F. (2004). *The Heavenly Writing: Divination, Horoscopy, and Astronomy in Mesopotamian Culture*. Cambridge University Press.

Schaeffer, R. et al. (2023). Are Emergent Abilities of Large Language Models a Mirage? *NeurIPS 2023*. https://arxiv.org/abs/2304.15004

Sharma, M. et al. (2024). Towards Understanding Sycophancy in Language Models. *ICLR 2024*. https://arxiv.org/abs/2310.13548

Shaw, I. (Ed.) (2000). *The Oxford History of Ancient Egypt*. Oxford University Press. (new pb ed. 2003)

Shleifer, A. & Vishny, R. W. (1997). The Limits of Arbitrage. *Journal of Finance*, 52(1). https://doi.org/10.1111/j.1540-6261.1997.tb03807.x

Shumailov, I. et al. (2024). AI Models Collapse When Trained on Recursively Generated Data. *Nature*, 631. https://doi.org/10.1038/s41586-024-07566-y

Soros, G. (1987). *The Alchemy of Finance*. Simon & Schuster.

Spennemann, D. H. R. (2025). "Delving into" the Quantification of AI-Generated Content on the Internet (Synthetic Data). *arXiv:2504.08755* [cs.IR], March 2025. https://arxiv.org/abs/2504.08755 [preprint, not peer-reviewed]

Stanford HAI (2026). *AI Index Report 2026*. Stanford University Human-Centered AI Institute, April 2026. https://hai.stanford.edu/ai-index/2026-ai-index-report

Taleb, N. N. (2012). *Antifragile: Things That Gain from Disorder*. Random House.

Taleb, N. N. & Sandis, C. (2014). The Skin in the Game Heuristic for Protection Against Tail Events. *Review of Behavioral Economics*, 1(1–2). https://doi.org/10.1561/105.00000006

Taleb, N. N. (2018). *Skin in the Game: Hidden Asymmetries in Daily Life*. Random House. Chinese translation: Zhou Luohua, trans., 《非对称风险》(in 《塔勒布经济学五部曲》[Taleb's Economics Quintet]), Beijing: CITIC Press, 2019. (In Chinese.)

Toda, A. A. (2023). 'Ergodicity Economics' Is Pseudoscience. arXiv:2306.03275. https://arxiv.org/abs/2306.03275 [preprint, not peer-reviewed]

Van De Mieroop, M. (1997). *The Ancient Mesopotamian City*. Oxford University Press. Chinese translation: Li Hongyan, trans., 《古代美索不达米亚城市》, Beijing: The Commercial Press, 2021. (In Chinese.)

van Dijk, J. (2000). The Amarna Period and the Later New Kingdom (c.1352–1069 BC). In I. Shaw (Ed.), *The Oxford History of Ancient Egypt*. Oxford University Press.

Wang, Defeng (2002). Bringing Historical Materialism Truly Onto the Stage: An Ontological Inquiry into the Concept of the "Real Individual." *Journal of Yunnan University (Social Sciences Edition)*, 1(1). (In Chinese.)

Wang, Defeng (2017). The Significance of *The Poverty of Philosophy* for Our Times. *Journal of Yunnan University (Social Sciences Edition)*, 16(6). (In Chinese.)

Wang, Defeng (2021a). *The Ideal and Fate of Classical Art* (lecture). (In Chinese.) Video: https://www.bilibili.com/video/BV1Cy4y137r1/

Wang, Defeng (2021b). *On Descartes* (lecture). (In Chinese.) Video: https://www.bilibili.com/video/BV1Pf4y1h7iJ

Wei, J. et al. (2022). Emergent Abilities of Large Language Models. *Transactions on Machine Learning Research*. https://arxiv.org/abs/2206.07682

Yu, H., Kim, D. & Kim, Y. (2026). Retrieval Collapses When AI Pollutes the Web. In *Proceedings of the ACM Web Conference 2026* (WWW '26). https://doi.org/10.1145/3774904.3792955

Zhuangzi. *Zhuangzi*, "The Primacy of Nourishing Life". Beijing: Zhonghua Book Company, 2010. (In Chinese.)

Zhuangzi. (2009). *Zhuangzi: The Essential Writings, with Selections from Traditional Commentaries* (B. Ziporyn, Trans.). Indianapolis: Hackett Publishing Company.